

Cluster-based healthcare network design problem with referral system using a hybrid genetic algorithm

Luqi Wang, Guoqing Yang*, Jianmin Xu

School of Management, Hebei University, Baoding, 071002, Hebei, China

ARTICLE INFO

Keywords:

Healthcare network design
County medical alliance
Cluster-based network
Referral system
Genetic algorithm

ABSTRACT

Addressing the unbalanced distribution of demands and medical resources is a particularly important issue in many healthcare systems. To achieve the equitable and efficient utilization of medical resources across regions, various medical alliances with tiered hospitals have been proposed and promoted to implement patient referrals. However, no formal analysis has been conducted on the implementation and management of medical alliances, especially over large geographical areas. This paper proposes the cluster-based healthcare network design problem with a referral system that provides a framework for integrating healthcare districting and patient referral problems within a hierarchical healthcare network design. It partitions the healthcare network into several clusters based on administrative features and designs diverse referral strategies for heterogeneous patients. To address the proposed problem, a mixed-integer linear programming model is formulated, and a hybrid genetic algorithm framework is developed to solve it efficiently. This algorithm considers the cluster-based nature of the healthcare networks and incorporates local search strategies to guarantee convergence performance. To demonstrate the efficiency of the proposed method, a case study is conducted involving 93 hospitals in Hebei, China. The results reveal that the proposed model can be extensively used to help decision-makers make informed decisions about constructing effective healthcare networks containing multiple medical alliances to reduce costs and improve efficiency. Furthermore, it suggests that a healthcare system equipped with a multi-hub configuration, diverse referral strategies, and a more relaxed capacity setting exhibits excellent performance in terms of costs and resilience. Finally, our study demonstrates that the proposed algorithm performs well in terms of efficiency and robustness.

1. Introduction

Human health and well-being are the key focuses of the United Nations' Sustainable Development Goals [1]. Despite worldwide efforts to reform healthcare systems, a significant gap remains between desired equity and efficiency. For instance, in China, a large portion of the population live in rural areas, while approximately 80% of medical resources are concentrated in urban hospitals. Furthermore, although primary hospitals account for 94.87% of all medical institutions, they shoulder only 54.16% of outpatient workload. By contrast, city hospitals, comprising a mere 3.78% of all medical institutions, bear the burden of 40.05% of the workload [2]. As the aging population increases, the country is under mounting pressure to address the conflict between the growing demand for more and better services and the inefficient use of medical resources.

Since 2017, the Chinese government has initiated healthcare reforms to implement County Medical Alliances (CMAs) to efficiently allocate healthcare resources and enhance the quality of grassroots

services [3]. CMAs provide a hierarchical structure within each county, typically consisting of county-level hospitals (CHs), township health institutions (THIs), and village medical centers (VMCs). As high-level hospitals, CHs are equipped with advanced medical resources and expertise and offer a comprehensive range of services. Primary hospitals (PHs), including THIs and VMCs, function as gatekeepers and serve patients with basic diagnoses or treatment services. Hospitals at each level are not separate entities and their cooperation is usually achieved through patient referrals. As a common healthcare delivery approach, rational referrals can decentralize patient visits, enhance the utilization of high-quality medical equipment, and improve the accessibility of medical services, thus benefiting patients. Recognizing these advantages, multiple cities in China have taken the initiative to implement CMAs. However, the layout of diverse healthcare facilities across multiple counties, along with the optimization of patient referral processes, poses significant challenges. Motivated by these challenges, we are dedicated to conducting research on the healthcare network

* Corresponding author.

E-mail address: yqg@hbu.edu.cn (G. Yang).

<https://doi.org/10.1016/j.seps.2025.102174>

Received 25 October 2023; Received in revised form 31 December 2024; Accepted 2 February 2025

Available online 11 February 2025

0038-0121/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

design problem (HNDP) with careful planning for patient referrals.

The HNDP involves decisions such as facility location, demand allocation, capacity setting, transportation, and patient flow planning. A well-designed healthcare network will help reduce expenses, enhance capacity utilization, and improve patient contentment, resulting in overall system efficiency improvement. An extension of the HNDP involves incorporating a districting problem. The districting problem aims to partition a geographical area into districts or clusters with the objective of enhancing management at the cluster level while achieving good system-wide performance [4]. Districting programs can be categorized into two groups based on their partition rules. One group is concerned with developing a partition plan that meets specific criteria such as compactness, contiguity, and balance [5,6]. This type of districting design has been discussed in domains such as public transportation, political organization, and healthcare services [7]. The second category focuses on defining each cluster based on realistic geographical and administrative features before making a decision [8,9]. This situation often happens in the real world, since control and management of local networks are heavily influenced by the associated administrative planning [10]. For instance, in China, the implementation of CMAs follows a cluster-based mode, where major cities are partitioned into multiple clusters, each tasked with managing a specific CMA. In this context, a well-designed cluster-based healthcare network is essential for advancing the nationwide development of CMAs. However, existing studies in the latter group have mainly focused on transportation, with little attention given to healthcare issues. To address this gap, we propose a novel cluster-based healthcare network design problem, in which the healthcare network is divided into clusters according to county divisions in advance, and subsequent network design decisions are constrained by these clusters.

Patient referral is a typical healthcare practice, enabling patients to obtain reasonable treatment at different hospital levels [11]. Many countries use hierarchical systems to achieve efficient referrals. In a two-tiered system, high-level hospitals provide services to all types of patients, while low-level hospitals focus on mild illnesses and refer severely ill patients to high-level hospitals [12]. However, this upward referral, together with patient preferences for high-quality services, has led to overcrowding in high-level hospitals and the underutilization of low-level hospitals. Thus, reverse referrals, that is referring patients from higher- to lower-level hospitals, play an important role in balancing resource utilization at the two levels [13]. Additionally, countries like China have adopted a two-way referral process, encompassing both upward and reverse referrals supported by dedicated policies and financial incentives. Despite these efforts, the absence of a standard referral mechanism, that is how to refer patients and what proportion of patients should be referred, hinders system efficiency and increases costs. Although there is a growing body of work discussing various referral strategies and the use of contracts to guide patient referrals [14], a gap remains in the consideration of patient conditions when designing referral policies. Additionally, the referral process between hospitals at the same level, often necessitated by variations in capabilities and capacities, has received limited attention. To address this gap, this study proposes a hybrid referral system (HRS) that categorizes patients based on their conditions and employs different referral strategies accordingly. Furthermore, referrals between high-level hospitals for severely ill patients are considered, aiming to enable patients to access optimal treatment promptly.

In this study, we propose the design of a cluster-based healthcare network with a referral system (CHNRS), in which each county is treated as a cluster with at least one CH and the other hospitals in this cluster as PHs. Each PH is assigned to a specific CH within its cluster and interconnections are established between the CHs across all clusters. To address the imbalance between CHs and PHs, the HRS is incorporated into the network. We develop a mixed-integer linear programming (MILP) model that minimizes the costs associated with healthcare network construction and patient referral within a

certain period. The decisions within this model involve determining the number and location of CHs in each cluster, allocating PHs to CHs, and optimizing patient flow under the HRS. To ensure that the proposed model can be effectively solved, we decompose it into two submodels under a heuristic framework. The first submodel, known as the network construction model (NCM), focuses on determining the healthcare network structure. The second submodel, referred to as the network flow model (NFM), optimizes the patient referral flow within the established healthcare network. We introduce a hybrid genetic algorithm (HGA) (also known as memetic algorithm) framework based on the cluster-based policy to solve the CHNRS problem. A local search (LS) procedure with two neighborhood structures is used to accelerate the solving process of the genetic algorithm. A case study in Hebei, China is conducted to assist local government in better planning the healthcare system in the context of CMAs.

In summary, the key contributions of our study are as follows:

- We innovatively adopt the cluster-based policy to capture real-world geographical and administrative features in healthcare network design. We go a step further by incorporating a hybrid referral system that considers the heterogeneity of patient conditions and provides diverse referral strategies for them. This study is the first to integrate healthcare network design, healthcare districting, and patient referrals, paving the way for enhanced efficiency and effectiveness in healthcare systems.
- We propose an MILP model that optimizes the CHNRS problem. To solve this model, we develop an HGA framework where the MILP model is decomposed into two submodels, NCM and NFM, to improve efficiency. Additionally, we consider the cluster-based nature when designing the chromosome structure and genetic operators of the HGA, ensuring a more effective optimization process.
- We conduct numerical experiments on a real case study of Hebei province, China to demonstrate the applicability of the proposed model and the efficacy of the developed solution approach. The results show that the proposed model can be applied as a flexible decision support tool for decision makers in designing effective healthcare networks, while also highlighting the efficiency of our solution algorithm, particularly when addressing large-scale instances.

The remainder of this paper is organized as follows. Section 2 provides a comprehensive overview of the related study. Section 3 provides a detailed description of the problem and model formulations. Our proposed HGA framework is presented in Section 4, followed by the numerical experiments in Section 5. Section 6 provides managerial insights. Finally, Section 7 presents the conclusions, limitations, and future avenues of this study.

2. Literature review

In this section, we review the literature in four separate but complementary research streams in the context of healthcare networks, including hierarchical healthcare network design, healthcare districting, patient referral, and solution methods.

2.1. Hierarchical HNDPs

HNDPs have gained growing importance in the success of healthcare systems because they can ensure that the objectives of the system design are served, including minimizing social costs or maximizing patients' benefits [15]. Numerous studies have been conducted on HNDPs. The goal here is not to review all the related works in the HNDP literature; rather, we direct the reader to exhaustive reviews of the topic provided by Rahman and Smith [16], Daskin and Dean [17], and Ahmadi-Javid et al. [18]. Most current healthcare systems are hierarchically structured because they consist of facilities at different

levels, such as clinics, health centers, and hospitals [19]. In such systems, a link often exists between different facility levels; thus, it is impossible to determine the location of each facility separately [20]. In this respect, Galvão et al. [21] proposed a hierarchical model for the location of maternal and perinatal healthcare facilities in the municipality of Rio de Janeiro and considered three main types of facilities that cooperated in a nested hierarchy with service referrals. Galvão et al. [22] extended the research of Galvão et al. [21] by considering the capacity constraints. Mestre et al. [23] presented a hierarchical multi-service network including district and central hospitals, where hospitals as multi-service suppliers can provide inpatient, outpatient, and emergency services. Zhang et al. [24] addressed a two-tier medical facility location problem, where community hospitals offer primary services, while general hospitals provide both primary and advanced diagnosis or treatment services. For an in-depth understanding of the hierarchical healthcare network design problem, refer to Farahani et al. [25], Ortiz-Astorquiza et al. [26], Karatas and Dasci [27], and Pourrezaie-Khaligh et al. [28].

2.2. Healthcare districting problem

Recently, an innovative stream has made exciting progress in incorporating the districting problem into the HNPD. Districting is the process of dividing a geographical area into small districts that represent service delivery units [29]. In the context of healthcare, the districting problem involves strategic decision-making that aims to identify service areas that include patients to be served by hospitals [4]. The output of the districting problem is a collection of service areas, called a district plan [30]. The district plan is not arbitrary but meets a series of planning criteria [7]. Several studies have considered predetermined upper or lower bounds on district size to achieve balanced districts [5]. Equitable workloads and travel times are popular balancing criteria in healthcare districting [31]. Compactness is another commonly used criterion to ensure minimal transportation or visit time within districts [6,32]. Other criteria, namely, contiguity, integrity, and connectivity, are also considered in healthcare districting [33]. Although existing studies have proposed various criteria and combinations to establish districting constraints, they have neglected realistic geographical and administrative features in healthcare network design. This aspect, also known as the cluster-based policy, has long been recognized in the field of transportation [8]. For instance, Yu et al. [9] and Shang et al. [10] have successfully incorporated the cluster-based policy into hub network systems by predetermining clusters (represented by geographical zones) for administrative and political reasons. These studies demonstrate that the cluster-based policy helps to align the constructed network with the area development, leading to better service in newly developed areas and overall system performance enhancement.

2.3. Patient referral problem

Patient referral is commonly used to address the imbalance between healthcare supply and demand across hierarchical hospitals [11]. Extensive research has explored diverse patient-flow patterns and developed efficient referral systems [34]. One branch of the literature focuses on the upward referral of patients, which involves transferring patients from lower-level medical facilities to higher-level hospitals for superior services [15,35]. Qiu et al. [36] implemented an intelligent upward referral from community health centers (CHCs) to general hospitals (GHs) for patients receiving upward referral admissions. Zhou et al. [37] used a Stackelberg game to investigate the impact of different medical care policies on upward patient flow. Other studies have often introduced a fixed ratio parameter into their optimization models to control for the number of upward referrals [24,38,39]. Another branch of the literature studies reverse referral, that is, diverting patients from upper- to lower-level hospitals, thereby balancing utilization at the

two levels and mitigating overutilization in top-tier hospitals [13]. Guo et al. [40] analyzed the referral incentives for compelling more patients to receive treatment in CHCs so that the overcrowding situation faced by most CHs can be alleviated. Wang et al. [41] incorporated patient revisits and coordination in reverse referrals, capturing equilibrium referring rates via a three-stage Stackelberg game with a queuing model. Additionally, Karakaya and Meral [42] and Huang et al. [43] highlighted the issue of two-way referrals between hospitals at different levels. Although various referral strategies, including upward, downward, and two-way referrals have been well discussed, a significant gap exists in the development of heterogeneous referral strategies for patients with diverse health conditions. Additionally, current research predominantly focuses on referrals between hospitals with different tiers without adequately addressing the real-world scenario of referrals occurring within hospitals at the same level.

2.4. Solution methods in HNDPs

Network design and districting problems are two well-known NP-hard problems [5,28], and most mathematical models in the HNPD literature have been in the form of mixed-integer linear programming. Researchers have primarily explored three types of techniques to find effective solutions to this problem: general-purpose optimization software, exact methods, and meta-heuristic algorithms. Commonly used commercial solvers include CPLEX [23,44] and GUROBI [27, 45]. Owing to the limited computing capacity of commercial solvers, many studies have attempted to solve the HNPD by employing relaxation techniques that involve relaxing integer constraints and working with non-integer solutions. Among them, the branch-and-bound algorithm [46,47] and more sophisticated methods, such as the Benders decomposition algorithm and branch-and-cut [15], can solve such problems to optimality for small- or medium-sized problem instances. However, despite their effectiveness and accuracy, these exact methods fail to solve large-scale real-world HNDPs. Consequently, most research efforts toward the development of algorithms for HNPD have focused on meta-heuristic algorithms, including Lagrangian relaxation [48], simulated annealing [49,50], tabu search [24,51], and genetic algorithms [52–54]. In particular, the genetic algorithm provides an attractive flexible framework for addressing network design problems, which can search for effective solutions in large and complex non-linear spaces [55].

2.5. Literature gaps

Table 1 summarizes the most important research conducted in the field of HNDPs, highlighting key features such as application, network structure, districting criteria, referral strategies, and solution approaches. The review of the literature reveals that the hierarchical healthcare network design, healthcare districting, and patient referral are often studied separately or partially integrated into HNDPs. This study aims to bridge this gap by simultaneously integrating all three practical topics, which, to the best of the authors' knowledge, have not been considered previously. In this study, the districting problem is based on a real-world application of CMAs, where each cluster is predetermined corresponding to the administrative division, that is, a county is a cluster. To the best of our knowledge, this partitioning criterion has not been implemented in healthcare research. Another significant gap lies in patient referral research, where referral rates are often set as fixed proportions and little attention is given to the heterogeneity of patient flow patterns under different healthcare conditions. Although a few studies have considered patient priorities, they have failed to develop differentiated referral strategies tailored to different types of patients. Additionally, previous studies have overlooked referrals between hospitals at the same level. Motivated by this, this study integrates a hybrid referral system (HRS) into the HNPD. In HRS, patients are stratified into two categories: mild patients (MPs)

Table 1
Summary of the work available in the literature.

Reference	Application	Hierarchical	Districting criteria					Referral					Solution approach
			Compactness	Contiguity	Balance	Integrity	Geographical	Upward	Downward	Two-way	Same-level	Patient heterogeneity	
Galvão et al. [19]	Healthcare	✓						✓					Benders decomposition
Galvão et al. [21]	Healthcare	✓						✓					Benders decomposition
Shumsky and Pinker [35]	Healthcare	✓						✓					Lagrangian heuristic
Boffey et al. [38]	Healthcare	✓						✓					Lagrangian heuristic
Harper et al. [31]	Healthcare				✓								Lagrangian heuristic
Galvão et al. [22]	Healthcare	✓						✓					CPLEX
Galvão et al. [39]	Healthcare	✓						✓					Tabu search
Yu et al. [9]	Transportation						✓						Genetic algorithm
Kim and Kim [46]	Healthcare												Genetic algorithm
Datta et al. [33]	Healthcare		✓			✓							
Benzarti et al. [32]	Healthcare		✓	✓		✓							
Steiner et al. [5]	Healthcare			✓		✓							
Qiu et al. [36]	Healthcare	✓						✓					
Zarrinpoor et al. [20]	Healthcare	✓						✓					
Li et al. [13]	Healthcare	✓							✓				Benders decomposition
Zarrinpoor et al. [15]	Healthcare	✓						✓					Lagrangian heuristic
Bidhandi et al. [50]	Healthcare												Genetic algorithm
Farughi et al. [29]	Healthcare		✓	✓									
Darmian et al. [4]	Healthcare			✓		✓							
Shang et al. [10]	Transportation	✓					✓						
Wang et al. [41]	Healthcare	✓							✓				
Zhang et al. [24]	Healthcare	✓						✓					GUROBI
Yu et al. [11]	Healthcare	✓						✓					Branch and bound
Zhou et al. [12]	Healthcare	✓						✓					Branch and bound
Karakaya and Meral [42]	Healthcare	✓								✓			Simulated annealing
Chouksey et al. [47]	Healthcare	✓						✓					Simulated annealing
Hao et al. [34]	Healthcare	✓								✓			Genetic algorithm
Huang et al. [43]	Healthcare	✓								✓			Genetic algorithm
Kummer et al. [54]	Healthcare									✓			Genetic algorithm
This paper	Healthcare	✓					✓			✓	✓	✓	Genetic algorithm

and severe patients (SPs). Distinctive referral strategies are specifically designed for each category. Moreover, patient referrals between CHs are considered in this system. This study addresses the challenges posed by two well-known NP-hard problems: the network design problem and the districting problem. The addition of patient referral decisions further increases the complexity of solving this problem. Many approaches proposed in the literature, including commercial solvers and exact methods, have limitations in handling large-sized instances. Therefore, owing to the proven advantages of the metaheuristic algorithm in managing complex computational tasks, particularly when dealing with substantial scale challenges, and its inherent flexibility to accommodate practical constraints and diverse objective functions, this study develops an HGA. In this algorithm, the chromosome structure and genetic operators are designed according to the cluster-based features of the healthcare network. The aim is to effectively address the intractability of the problem and efficiently solve large-sized instances, providing a robust solution approach.

3. Problem description and model formulation

3.1. Problem description and assumption

In this study, we focused on the CHRNS problem. The upper layer of the healthcare network connects all CHs, forming a complete network structure. The lower layer integrates the allocation of PHs to their CHs in each cluster using a star structure. The healthcare network is defined as an undirected graph $G(N, A)$, where N denotes the set of all hospital nodes categorized into hub nodes (CHs) and non-hub nodes (PHs), and A represents the set of all connections between hospitals at both levels. The network comprises K clusters based on the cluster-based policy, where N_k denotes the set of all nodes in the k th cluster. To construct this network, at least one node i must be selected to build a CH in each cluster, incurring a construction cost F_i and a maximum capacity Q'_i . The remaining nodes constitute the PHs, each corresponding to a lower service capacity Q . Each PH should be assigned to a CH in its cluster, and by default, a two-way link exists between any two CHs. To provide an explicit depiction of the network structure, we present a concrete illustration of a network with 4 clusters, 5 hubs, and 15 non-hubs in Fig. 1.

Once the healthcare network is established, we can determine the optimal flow of patients. Our hybrid referral system considers that the number of MPs and SPs, and the resources they consume, vary. Patients in non-hub nodes have the option to receive treatment locally or visit the CH they belong to. Patients in hub nodes can opt to remain in

local CHs, regardless of whether they are MPs or SPs, and MPs may be recommended to seek treatment at a subordinate PH. In particular, SPs within hub nodes may be referred to other CHs in any cluster. To avert hospital congestion and the potential collapse of high-level hospitals, all patient referrals should be implemented within the capacity limits of the receiving hospitals. Fig. 2 provides a clear depiction of the concrete referral options available. Furthermore, as the cost of patients is a continuous input in the medical system, we accounted for a time period factor and considered one year as a time period.

This study aimed to address three types of decisions: location decisions (selection of CHs), assignment decisions (allocation of PHs), and referral decisions (patient flow between hospitals). The primary objective was to achieve the lowest construction and connection costs, followed by the secondary objective of minimizing the total treatment cost, including treatment and referral costs. To achieve these objectives, we propose an MILP model. For convenience of modeling, we make several assumptions:

- The nodes within the healthcare network symbolize hospitals, and simultaneously, each node represents local medical demand.
- Clustering rules have been predetermined such that each county forms a cluster.
- Patients within a given node can receive medical treatment locally or at a CH within their respective cluster. Visits to PHs in other clusters are not permitted.
- Although the overall number of CHs and PHs is known, the precise number of hospitals in each region remains undetermined.
- The maximum service capacity for both types of hospitals is well-established.

3.2. Model formulation

3.2.1. Notations

All the sets, parameters, and decision variables are defined as follows.

Sets:

- N : The set of all nodes;
- K : The set of all clusters;
- N_k : The set of all nodes in the k_{th} cluster;

Parameters:

- F_i : Fixed cost of opening a CH at node i ;

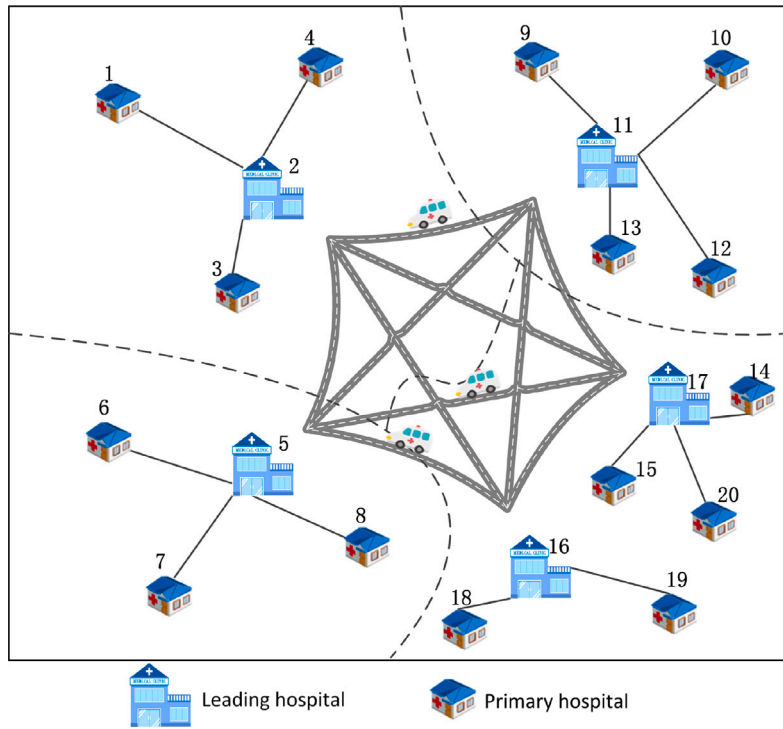


Fig. 1. An illustration of the cluster-based healthcare network.

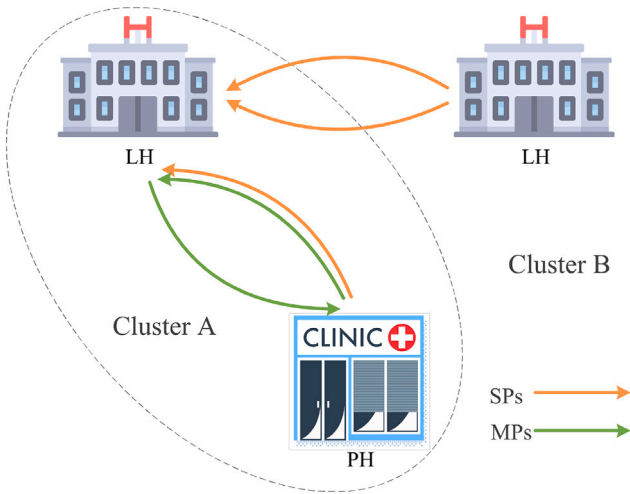


Fig. 2. An illustration of referral strategies in the healthcare network.

C_{ij} : The cost of constructing the link between hospitals i and j ;
 c_i : The cluster index which node i belongs to;
 β : The factor by which the construction cost of the link between hubs is multiplied when compared to the link between hubs and non-hubs;
 d_i^s : Number of patients with type s_{th} disease in node i ;
 C_{ij}^s : The cost for the patient with s_{th} disease at node i to visit the doctor at node j ;
 λ_s : Unit medical resources required to treat a patient with s_{th} disease;
 Q_i : Maximum service capability when node i is a PH;
 Q_i' : Maximum service capability when node i is a CH;
 L : Number of time periods;
 M : A large positive constant;

Decision variables:

x_{ij} : 1, if node i is assigned to node j ; 0, otherwise; note that x_{ii} equals 1 if a CH is opened at node i ; 0, otherwise;
 y_{ij} : 1, if nodes i and node j ($i \neq j$) are both CHs; 0, otherwise;
 z_{ij} : Number of patients with the first type of disease at node i visiting a doctor at node j ;
 m_{ij} : Number of patients with the second type of disease at node i who see the doctor at node j ;

3.2.2. Model formulation for the CHNRS problem

This subsection focuses on the formulation of the CHNRS problem. The primary decision-making concerns the selection of CHs, allocation of PHs, and patient referral flow within the established healthcare network. Our primary goal was to reduce the costs associated with constructing the healthcare network, while minimizing patients' medical expenses. We present the model as follows:

$$\min \sum_{i \in N} F_i x_{ii} + \left(\sum_{i \in N} \sum_{j \in N} C_{ij} x_{ij} + \sum_{i \in N} \sum_{j \in N, j \leq i} \beta C_{ij} y_{ij} \right) + \sum_{i \in N} \sum_{j \in N} L (C_{ij}^1 z_{ij} + C_{ij}^2 m_{ij}) \quad (1)$$

$$\text{s.t.} \quad \sum_{i \in N_k} x_{ii} \geq 1, \forall k \in K \quad (2)$$

$$\sum_{j \in N_k} x_{ij} = 1, \forall i \in N_k, k \in K \quad (3)$$

$$x_{ij} = 0, \forall i, j \in N, c_i \neq c_j \quad (4)$$

$$x_{ij} \leq x_{jj}, \forall i, j \in N \quad (5)$$

$$y_{ij} \geq x_{ii} + x_{jj} - 1, \forall i, j \in N, i \neq j \quad (6)$$

$$2y_{ij} \leq x_{ii} + x_{jj}, \forall i, j \in N, i \neq j \quad (7)$$

$$\sum_{j \in N} z_{ij} = d_i^1, \forall i \in N \quad (8)$$

$$\sum_{j \in N} m_{ij} = d_i^2, \forall i \in N \quad (9)$$

$$z_{ij} \leq M(x_{ij} + x_{ji}), \forall i, j \in N, i \neq j \quad (10)$$

$$m_{ij} \leq M(x_{ij} + y_{ij}), \forall i, j \in N, i \neq j \quad (11)$$

$$\sum_{i \in N} (\lambda_1 z_{ij} + \lambda_2 m_{ij}) \leq (1 - x_{jj})Q_j + x_{jj}Q'_j, \forall j \in N \quad (12)$$

$$x_{ij}, y_{ij} \in \{0, 1\}; z_{ij}, m_{ij} \in R^+, \forall i, j \in N \quad (13)$$

Objective function (1) aims to minimize the total cost incurred over L periods and is composed of three terms. The first term focuses on reducing the cost of opening CHs. The second term minimizes the connection cost, encompassing the resource and technique flows between CHs and between CHs and PHs. The final term aims to minimize total patient treatment costs. Constraint (2) limits each cluster to have at least one CH. Constraint (3) assigns each PH to a single CH within its cluster, while Constraint (4) prohibits the allocation of a PH to an CH in other clusters. Constraint (5) guarantees that no hospitals are assigned to a PH. In other words, if $x_{ij} = 1$ for $i \in N$, then $x_{jj} = 1$ must hold true. Constraints (6)–(7) are the auxiliary linear constraints. Constraints (8)–(9) ensure that all patients within each cluster can be served. Constraints (10)–(11) dictate that patients are allowed to be referred from hospital i to hospital j if there is cooperation between these two hospitals. These two constraints can be reformulated more tightly by replacing the big M parameter with system capacities. Constraint (12) ensures that the maximum service capacity of each hospital is satisfied. Finally, Constraint (13) specifies the domains of the decision variables.

3.3. Model analysis

The CHNRS problem was formulated as an MILP model. In this section, we present a comprehensive analysis of the MILP model from two perspectives. First, we examined the impact of Constraint (2) on the healthcare network structure, which limits the number of hubs within each cluster. Second, we analyzed the solving complexity of the MILP model and decomposed it into two submodels: NCM and NFM. Subsequently, we present an improved NFM in which the number of variables was significantly reduced.

Most cluster-based network design studies have assumed a single-hub form for each cluster. However, in the healthcare context, each cluster commonly has multiple hubs. For instance, when implementing CMAs, several hospitals such as county hospitals, Chinese medicine hospitals, and maternal and childcare centers, are often selected as CHs within a county. Therefore, this study extended the healthcare network to a multi-hub form using Constraint (2). By modifying Constraint (2) from $\sum_{i \in N_k} x_{ii} \geq 1$ to $\sum_{i \in N_k} x_{ii} = 1$, the healthcare network can degenerate to a single-hub form. Although the multi-hub form is more complex to handle than the single-hub form, it offers distinct advantages in designing an effective healthcare network, as demonstrated in Section 5.

The CHNRS problem can be viewed as a variant of the facility location or network design problem, both of which are NP-hard combinatorial optimization problems [28,56]. Therefore, even without considering the complexity introduced by the cluster-based structure and referral system, the MILP model inherently falls within the NP-hard category, which makes it difficult to achieve an exact solution within a limited computational time. Consequently, recourse to heuristic algorithms is imperative, particularly for addressing extensive problem scales. However, given the large number of two-dimensional ($N \times N$) real-valued referral variables involved and the capacity constraints imposed on them, it is difficult to obtain optimal solutions for the patient referral problem under a specific healthcare network structure. Driven by this, we decomposed the MILP model into two submodels under a heuristic solving framework: NCM and NFM. The NCM involves decisions about selecting CHs and allocating PHs to CHs with the goal of reducing healthcare network construction costs. The NFM optimizes patient flow with the established healthcare network with the objective of minimizing patient treatment costs. The NCM is expressed as follows:

Table 2
Notations for the INFM model.

Symbol	Description
<i>Sets</i>	
N_B	The set of all non-hub nodes
N_H	The set of all hub nodes
N_i	The set of all subordinate nodes of hub node i (including node i itself)
<i>Parameters</i>	
σ	The multiplier factor of the resource consumption of treating a SP to a MP
h_i	The label of the upper-level hub to which node i belongs
<i>Variables</i>	
z_i	Number of local MPs in node i who remain in the local area for treatment
z'_i	Number of MPs who visit the higher level CH to which node i belongs
m_i	Number of local SPs in node i who remain in the local area for treatment
m'_i	Number of SPs who visit the higher level CH to which node i belongs
x_i	Number of MPs received by node i who were transferred from CHs
y_{ij}	Number of SPs from CH i visiting CH j

$$(\text{NCM}) \min \sum_{i \in N} F_i x_{ii} + \sum_{i \in N} \sum_{j \in N} C_{ij} x_{ij} + \sum_{i \in N} \sum_{j \in N, j \leq i} \beta C_{ij} y_{ij} \quad (14)$$

$$\text{s.t.} \quad \sum_{i \in N_k} x_{ii} \geq 1, \forall k \in K \quad (15)$$

$$\sum_{j \in N_k} x_{ij} = 1, \forall i \in N_k, k \in K \quad (16)$$

$$x_{ij} = 0, \forall i, j \in N, c_i \neq c_j \quad (17)$$

$$x_{ij} \leq x_{jj}, \forall i, j \in N \quad (18)$$

$$y_{ij} \geq x_{ii} + x_{jj} - 1, \forall i, j \in N, i \neq j \quad (19)$$

$$2y_{ij} \leq x_{ii} + x_{jj}, \forall i, j \in N, i \neq j \quad (20)$$

$$x_{ij}, y_{ij} \in \{0, 1\}, \forall i, j \in N, \quad (21)$$

where the notations for the sets, variables, and parameters remaining the same as those in model (1). When the NCM determines the healthcare network structure, variables x and y are transformed into parameters of the NFM. Thus, the NFM is formulated as follows:

$$(\text{NFM}) \min \sum_{i \in N} \sum_{j \in N} L(C_{ij}^1 z_{ij} + C_{ij}^2 m_{ij}) \quad (22)$$

$$\text{s.t.} \quad \sum_{j \in N} z_{ij} = d_i^1, \forall i \in N \quad (23)$$

$$\sum_{j \in N} m_{ij} = d_i^2, \forall i \in N \quad (24)$$

$$z_{ij} \leq M(x_{ij} + x_{ji}), \forall i, j \in N, i \neq j \quad (25)$$

$$m_{ij} \leq M(x_{ij} + y_{ij}), \forall i, j \in N, i \neq j \quad (26)$$

$$\sum_{i \in N} (\lambda_1 z_{ij} + \lambda_2 m_{ij}) \leq (1 - x_{jj})Q_j + x_{jj}Q'_j, \forall j \in N \quad (27)$$

$$z_{ij}, m_{ij} \in R^+, \forall i, j \in N. \quad (28)$$

The application of heuristic algorithms in NCMs is well-established. Moreover, the NFM belongs to the category of linear programming (LP) models that can be efficiently solved using LP solvers. Therefore, decomposing the MILP model provides an opportunity to employ heuristic algorithms effectively. In other words, we can explore each potential healthcare network structure using heuristic algorithms, and

Table 3
Comparison of the number of variables in NFM and INFM.

N	NFM	$\alpha = 1$		$\alpha = 1/2$		$\alpha = 1/4$	
		INFM	Δ NFM (%)	INFM	Δ NFM (%)	INFM	Δ NFM (%)
20	800	500	37.5	200	75.0	125	84.4
50	5000	2750	45.0	875	82.5	406	91.9
80	12 800	6800	46.9	2000	84.4	800	93.8
110	24 200	12 650	47.7	3575	85.2	1306	94.6

subsequently solve the corresponding referral problem by calling computational solvers, thereby deriving feasible solutions for the MILP model. In addition, because of the sequential nature of the decomposed models during solution, where the network structure is determined first, followed by optimizing patient referrals under that structure, it becomes possible to transform two-dimensional referral variables into one-dimensional variables. As a result, the NFM is formulated as an improved model, hereafter referred to as the improved network flow model (INFM), with a reduced number of variables. Using the notations from the MILP model and the additional notations listed in Table 2, we present the INFM as follows.

$$(\text{INFM}) \min \sum_{i \in N_B} (z_i c_{ii} + z'_i c_{ih_i} + m_i c'_{ii} + m'_i c'_{ih_i}) + \sum_{i \in N_B} x_i c_{hi} + \sum_{i \in N_H} \left(\sum_{j \in N_H} y_{ij} c'_{ij} + x_i c_{ii} \right) \quad (29)$$

$$\text{s.t. } z_i + z'_i = d_i^1, \forall i \in N_B \quad (30)$$

$$m_i + m'_i = d_i^2, \forall i \in N_B \quad (31)$$

$$\sum_{j \in N_i} x_j = d_i^1, \forall i \in N_H \quad (32)$$

$$\sum_{j \in N_H} y_{ij} = d_i^2, \forall i \in N_H \quad (33)$$

$$z'_i = x_i, \forall i \in N_H \quad (34)$$

$$m'_i = y_{ii}, \forall i \in N_H \quad (35)$$

$$z_i + \sigma m_i + x_i \leq Q_i, \forall i \in N_B \quad (36)$$

$$\sum_{j \in N_i} (z'_i + \sigma m'_j) + \sum_{j \neq i \in N_H} \sigma y_{ji} \leq Q'_i, \forall i \in N_H \quad (37)$$

$$z_i, z'_i, m_i, m'_i, x_i, y_{ij} \in R^+, \forall i, j \in N \quad (38)$$

The objective function and constraints in this model have the same meaning as in the NFM. However, the model's presentation was altered because the two-dimensional variables, z and m , which describe patient referrals, were transformed into five one-dimensional variables. Additionally, a new two-dimensional variable, y , was introduced to describe the referral flow between the CHs.

Let $|N|$ denote the total number of hospitals in the network and α represent the proportion of CHs among all the hospitals. The NFM model has the variable size of $2|N|^2$, while the INFM model has a variable size of $5|N| + (\alpha|N|)^2$. Because the number of CHs was not predetermined, the value of α cannot be precisely determined. Therefore, we listed the variable improvements for α values of 1, 1/2, and 1/4 in Table 3. The results demonstrate a variable reduction of 40% to 50% when α has a maximum value of 1, which is not feasible in practice. For $\alpha = 1/4$, the variable reduction percentage exceeds 90%. Additionally, because α is typically much less than 1/4 in practice, the percentage reduction in the number of variables was even more significant. The improved model exhibited remarkable superiority.

4. Solution method

The CHNRS comprises two well-known NP-hard problems: the network design problem and the districting problem [5,28]. Moreover, incorporating patient referrals increases the problem's complexity, resulting in a larger number of variables and a more complex structure.

Therefore, as the problem size increases, exact methods become inefficient and time-consuming. In such cases, metaheuristic methods offer high efficiency in tackling complex optimization problems [57]. Among the metaheuristic algorithms, the genetic algorithm (GA) is one of the most popular and extensively studied approaches across various domains, showcasing its effectiveness and robustness. The GA offers several advantages over other evolutionary algorithms. First, the code of decision variables helps simulate the processes of biological genes, chromosomes, and genetic evolution, facilitating the representation of solutions and application of genetic operators [58]. For instance, in this study, GA can effectively incorporate cluster-based features into the chromosome representation and preserve this feature throughout the evolutionary process. Secondly, the GA's group search characteristics ensure population diversity, avoiding premature convergence toward local optimal solutions. It delves into the global and local search dimensions, enabling a comprehensive and effective exploration of the solution space for discovering high-quality solutions [59]. Moreover, the GA exhibits strong scalability and adapts well to different problem sizes. It efficiently explores larger search spaces without significant performance degradation. Additionally, the GA can be easily integrated with other technologies, allowing researchers to combine optimization techniques or problem-specific algorithms to create high-performance algorithms tailored to specific problem domains [60]. For example, local search techniques are commonly integrated into genetic algorithms to enhance their performance. These hybrid genetic algorithms, often referred to as memetic algorithms [61], represent one of the recent growing areas of research in evolutionary computing. In this study, the GA was selected to solve the proposed model. To improve its performance, we proposed the HGA, which combines a GA with a LS procedure and elitist strategies. As a consequence of incorporating these strategies, the convergence characteristic of the HGA is enhanced and its convergence process is significantly expedited. Further, we develop an algorithm framework by combining the HGA with an LP solver. This section begins by introducing the algorithm framework and subsequently delves into the details of the HGA.

4.1. Algorithm framework

As mentioned above, the MILP model was divided into two submodels: the NCM, which determines the healthcare network structure, and the NFM, which controls for patient referrals after the healthcare network is established. We establish an algorithm framework to manage these two submodels. In this framework, when generating a solution, a genetic algorithm combined with local search strategies (referred to as HGA) was employed to generate the feasible solution of the NCM. Subsequently, the improved NFM was solved by calling the CPLEX solver in HGA. By adding the objectives of the NCM and NFM, we ultimately obtained a feasible complete solution for the MILP model. The evolutionary process follows a similar procedure for each solution. Notably, solving the NCM and NFM within this algorithm framework is equivalent to solving the complete MILP using the HGA because both methods find suboptimal solutions by generating and evolving feasible solutions using specific genetic operators. The main difference is that solving the NCM and NFM generates a complete solution in two steps each time, which reduces the complexity of managing a large number of real-valued referral variables using a heuristic method and improves the efficiency of the solution process.

	N1				N2				N3				N4							
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
HubArray	0	1	0	0	1	0	0	0	0	0	1	0	0	0	0	1	1	0	0	0
AssignArray	2	2	2	2	5	5	5	5	11	11	11	11	11	17	17	16	17	16	16	17

Fig. 3. An example of the chromosome.

4.2. HGA algorithm

It is widely recognized that the genetic algorithm effectively conducts a global search across all feasible search spaces; however, it sometimes falls short in exploring the search space within the convergence area. To bolster the precision and convergence of the GA, we introduced an LS procedure featuring two neighborhood structures.

The performance of the GA is largely influenced by the quality of the initial solution. We applied the LS procedure to every individual generated during the initialization process, ensuring a promising start. Additionally, the LS procedure is helpful in accurately identifying superior local solutions, thereby significantly accelerating the exploration process. Therefore, it is embedded within the crossover and mutation operators to search the neighborhoods whenever a new individual is generated by either of these genetic operators. Furthermore, we established several probabilistic parameters to modulate the LS intensity, thereby preventing the algorithm from trapping in a local optimum. The pseudo-code for the HGA is provided in Algorithm 1.

In the following sections, we describe the main components of the HGA, including solution representation, population initialization, evolutionary process, and the LS procedure.

4.2.1. Solution representation

Chromosomes are typically represented as data structures by arrays. Each chromosome consists of two one-dimensional arrays: Hubarray and Assignarray. The Hubarray was used to describe the hub selection within the network and comprises variables with values of 0 or 1. A value of 1 indicates that the corresponding node is selected as a hub node, while a value of 0 signifies that it is a non-hub node. Assignarray depicts the assignments from non-hub nodes to hub nodes. If the value at position i of an Assignarray is m , node i is assigned to the hub node m . Notably, a hub node is self-assigned, meaning that its corresponding value in Assignarray is its serial number.

Our algorithm differs from the conventional chromosome structure in that it divided the chromosome into separate segments to accommodate the cluster-based nature of healthcare networks. Each segment in Hubarray and Assignarray corresponds to one cluster within the healthcare network. An illustration of the chromosome structure corresponding to the healthcare network in Fig. 1 is presented in Fig. 3. Because the network was divided into four clusters, this chromosome comprised four segments.

Note that the healthcare network structure represented by such a chromosome does not guarantee the feasibility of a lower-level NFM. In cases where the network structure lacks an adequate number of CHs, it may fail to meet the demands of all patients. The presence of solutions that are solely feasible for NCM can significantly reduce the efficiency of the algorithm because substantial time is consumed exploring infeasible regions or identifying infeasible solutions. To mitigate this issue, we incorporated patient information into the construction of the healthcare network and implemented a prejudgment procedure. The specific steps of the prejudgment procedure for a given chromosome are as follows:

1. For a new chromosome i , calculate the total quantity of medical resources needed by patients at each PH that exceeds its service

capacity and requires external transfers. This quantity is denoted as $O_{p,i}$, and $O_{p,i} = \sum_{n \in N_p} (d_n^1 + \lambda_2 d_n^2) - Q_n$, where N_p is the set of PHs, n represents a PH, and the meaning of other symbols are described in Section 3.2.1.

2. Calculate the total quantity of medical resources required by patients at all CHs. This quantity is denoted as $O_{l,i}$, and $O_{l,i} = \sum_{n \in N_l} (d_n^1 + \lambda_2 d_n^2)$, where N_l is the set of CHs, n represents an CH, and the meaning of other symbols are described in Section 3.2.1.
3. Calculate the sum of the maximum capacities of all CHs, and denote this total as $S_{Q_{l,i}}$.
4. If $O_{p,i} + O_{l,i} > S_{Q_{l,i}}$, then the referral problem associated with chromosome i is declared infeasible, resulting in invalidation of chromosome i ; otherwise, chromosome i is considered feasible.

This procedure is executed whenever a new chromosome is generated, whether during the initialization, the evolutionary process, or as a result of the LS procedure. If an infeasible chromosome is generated during the initialization process, it is repeatedly regenerated until a feasible chromosome is obtained. However, if an infeasible chromosome is generated during the evolutionary process, it is discarded, and the feasible parent chromosome is retained.

4.2.2. Population initialization

To start the GA, the first step involves generating an initial population. Typically, the initial population for the GA is generated using a random process or heuristic methods. The former ensures population diversity, while the latter provides high-quality initial solutions. In this study, we employed a two-step approach to generate the initial population, initially utilizing a random method, followed by application of the LS procedure with a specified probability. This two-step process aims to enhance the quality of the initial individuals and enable the GA to commence iterations with a favorable start.

Operationally, we began by defining a probability parameter P_i . For each node i within segment k , we determined whether it should be designated as a hub node based on a comparison between a randomly generated number between 0 and 1 and the value of P_i . If the random number was less than P_i , node i was assigned as the hub node; otherwise, it was not assigned. Once each position within segment k of the Hubarray was assigned a value, we examined this segment of the chromosome to verify the presence of at least one hub node. If this requirement was satisfied, the generated segment was retained. However, if the requirement was not fulfilled, the aforementioned process was repeated until a satisfying segment was obtained. By repeating these steps for each segment, we obtained an initialized Hubarray. Notably, the parameter P_i was adjustable, and its value increases when the capacity constraints of the healthcare network under consideration are more stringent. This adjustment compensates for situations in which additional CHs are required to meet patient needs. Conversely, when the capacity constraints are less restrictive, parameter P_i can be appropriately decreased.

After determining the hub quantity and locations, assignments from non-hubs to hubs were established. If an interval contained only one hub i , all non-hubs within the same interval were assigned to this hub node, resulting in all values within the interval of Assignarray being set to i . When multiple hubs were present within an interval, we randomly selected one hub for each non-hub as the parent hub. Furthermore, all hubs were assigned to themselves. Following initialization of the

Algorithm 1 The HGA pseudocode.

Require: All parameters
Ensure: Optimal solution

```

1:  $gen \leftarrow 0$ ;
2: Initialize  $P(gen)$ ;
3: Do LS on the individuals in  $P(gen)$  and replace  $P(gen)$  with  $P_1(gen)$ ;
4: while (termination criterion is not satisfied) do
5:    $gen \leftarrow gen + 1$ ;
6:   Generate  $P(gen)$  from  $P_1(gen - 1)$  by selection;
7:   if  $rand(0, 1) < p_c$  then
8:     Do one-point crossover and then LS for individuals in  $P(gen)$  and update  $P(gen)$  to  $P_2(gen)$ ;
9:   else
10:    Do two-point crossover and then LS for individuals in  $P(gen)$  and update  $P(gen)$  to  $P_2(gen)$ ;
11:   end if
12:   if  $rand(0, 1) < p_m$  then
13:     Do two-exchange mutation and then LS for individuals in  $P_2(gen)$  and update  $P_2(gen)$  to  $P_1(gen)$ ;
14:   else
15:     Do site mutation and then LS for individuals in  $P_2(gen)$  and update  $P_2(gen)$  to  $P_1(gen)$ ;
16:   end if
17:   Return the best solution in  $P_1(gen)$ ;
18: end while

```

Hubarrays and Assignarrays, we obtained a set of chromosomes for the NCM. By solving the NFM for each chromosome and aggregating the objective values of the NCM and NFM, we obtained the entire solution to the CHNRS problem.

4.2.3. Selection process

The selection operator is a critical component in GAs, Responsible for choosing individuals with higher fitness as parents for reproduction and inheritance, thereby guiding population evolution. Among various selection algorithms, roulette wheel selection is the most common. Its principle is to determine selection probabilities based on an individual's fitness relative to others in the population, giving those with higher fitness a greater chance of being chosen. Roulette wheel selection offers advantages in terms of simplicity and adherence to evolutionary principles. First, it preserves individuals with higher fitness, increasing their likelihood of entering the next generation and passing down their favorable traits. Second, it helps prevent the algorithm from becoming trapped in local optima. Even individuals with lower fitness have a chance of being selected, although their selection probability is lower. This allows the potential to overcome the local optima and enhances the algorithm's global search capability. It is worth mentioning that, in our research, individuals with smaller objective values were considered superior to those with larger values.

4.2.4. Crossover operator

The crossover operator generates offspring chromosomes by recombining the genes of the two parent chromosomes. This process ensures that the offspring inherit the characteristics of their parents. In our study, we employed two different crossover operators, namely one-point crossover and two-point crossover, to address the hub location and non-hub allocation problems. Unlike conventional crossover operators, our approach considered not only the entire chromosome but also each segment within the chromosome. To control the crossover process, we introduced a preset probability parameter p_c . The crossover was performed only when a randomly generated number between 0 and 1 was smaller than p_c . We now provide a detailed description of the steps involved in the crossover procedure.

Initially, two distinct chromosomes were randomly selected from the current population to serve as parent individuals. To perform one-point crossover, we randomly determined a crossover point for each of the two parent chromosomes. Subsequently, the segments of the Hubarray and Assignarray beyond this crossover point were exchanged. It is important to note that this type of crossover operator may occasionally generate offspring chromosomes that fail to meet the required conditions, specifically the absence of hubs within the interval

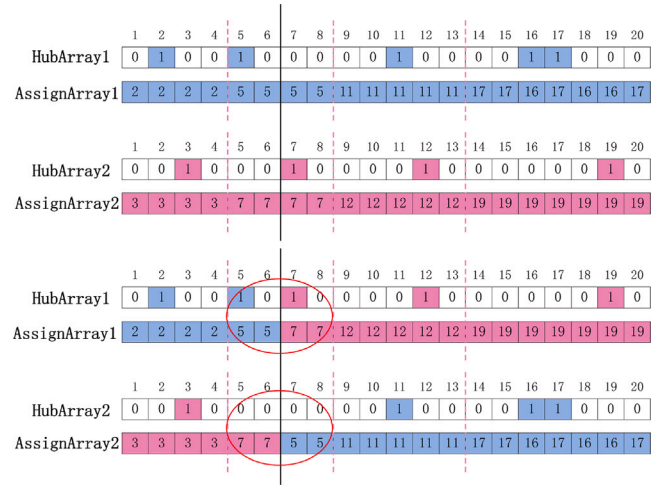


Fig. 4. An illustration of the one-point crossover operator.

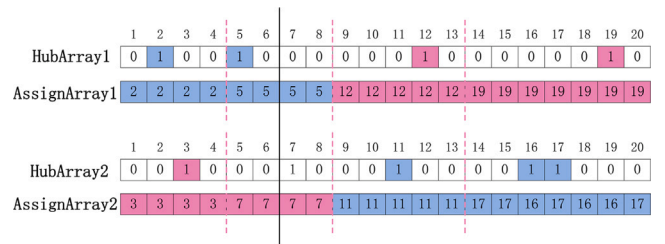


Fig. 5. A repaired offspring chromosome.

where the crossover point is located. To address this issue, a repair procedure was implemented for the offspring chromosomes. If a given interval lacked a hub after crossover, we retained the original hub and assigned all nodes within that interval to this hub. Using this approach, we obtained two feasible offspring chromosomes. By repeating the crossover process pop_size times, we generated an evolved population. Fig. 4 illustrates an example of the one-point crossover process. The hub nodes of the first parental chromosome were 2, 5, 11, 16, and 17, while the hub nodes of the second parental chromosome were 3, 7, 12, and 19. The crossover point was situated within the second interval. After the initial crossover step, the second chromosome lacked hubs in

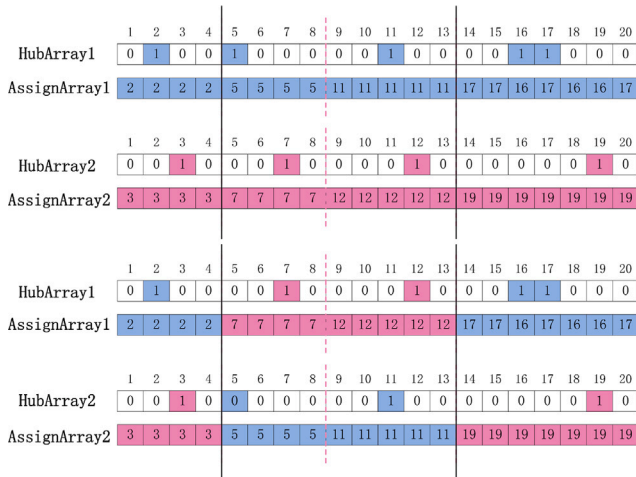


Fig. 6. An illustration of the two-point crossover operator.

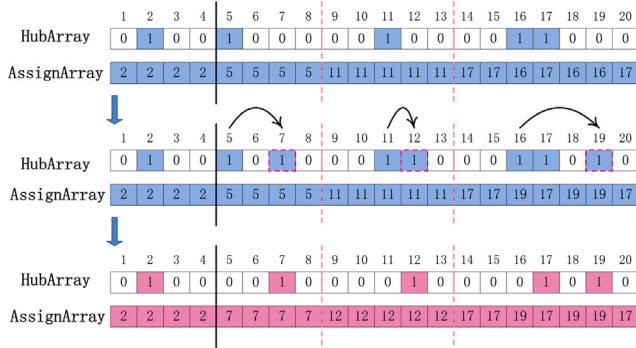


Fig. 7. An illustration of the two-exchange mutation operator.

its second interval, violating the required condition. Consequently, a repair procedure was employed to obtain a pair of feasible solutions, as shown in Fig. 5.

Similar to the one-point crossover, the two-point crossover involves the exchange of segments of genes between parent chromosomes. The key distinction lies in the number and selection methods of crossover points. The fundamental concept is to randomly select two distinct points, denoted by a and b , and exchange the segments between them. However, this approach for selecting crossover points can potentially result in offspring chromosomes that violate the required conditions. To address this issue, we adopted a modified approach in which we first randomly chose two intervals and considered the right boundaries, a and b , of these intervals as crossover points. Subsequently, we exchanged the segments between a and b of the two chromosomes, thereby generating two new feasible chromosomes. A detailed depiction of the two-point crossover operator is presented in Fig. 6.

4.2.5. Mutation operator

The mutation operator is essential for maintaining population diversity and preventing premature convergence. Unlike crossovers, which recombine existing genetic information, mutations can create entirely new variations by altering gene values or introducing new genes. This exploration capability allows the algorithm to overcome local optima and obtain globally optimal solutions. Considering the structural characteristics of chromosomes, two types of mutation operators were employed: two-exchange and site mutations. Both methods are based on a single mutation. Additionally, a predetermined mutation probability parameter, denoted by p_m , controlled the mutation rate.

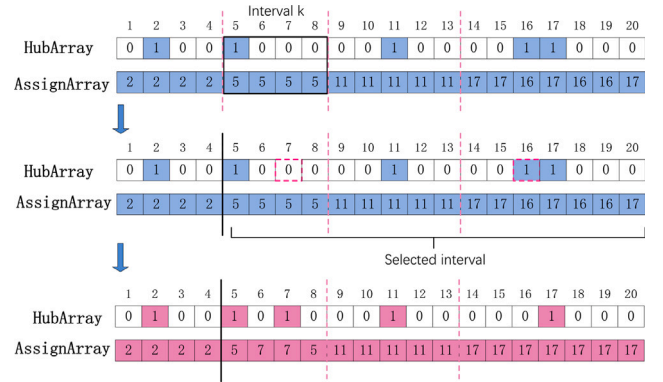


Fig. 8. An illustration of the site mutation operator.

To perform the two-exchange mutation operator, we first generated a random integer k within the range $[1, K]$, where K denotes the number of clusters in the healthcare network. Once we selected cluster k , the mutation operator was applied to adjacent intervals, either prior to or subsequent to the chosen cluster. In each mutation interval, we randomly selected a non-hub node and modified its value from 0 to 1. The original hub node value was set to 0. By repeating this procedure for all selected intervals, a new feasible offspring chromosome was obtained. A detailed depiction of the two-exchange mutation operator is shown in Fig. 7.

The two-exchange mutation operator alters only the hub location within each cluster and does not change the number of hubs. By contrast, the site-mutation operator focuses on modifying the number of hubs in each cluster. To perform this mutation, a random interval k is selected within the parent chromosome, and the section following this interval is selected for the mutation. Within the selected intervals, nodes were randomly chosen. Non-hub nodes were transformed into hub nodes, and some non-hub nodes were assigned to them. If the selected node is already a hub node (in cases where the interval contains more than one hub), it is converted into a non-hub node and its subordinate nodes are assigned to other hubs. During this process, various factors such as treatment costs, connection costs, and other relevant information are considered for node reassignment. The degree of randomness was maintained throughout the assignment process to introduce variability. Fig. 8 provides a detailed depiction of the site mutation operator.

4.2.6. Elitist strategy

As an inherent feature of GAs, elitism plays a crucial role in mitigating genetic drift by preserving the best-performing individual and allowing its attributes to be passed/copied to the next generation. Elitism can enhance selection pressure by eliminating low “salient” genes from chromosomes, which results from insufficient selection pressure. This, in turn, often improves the optimality and convergence performance of GAs. In our proposed algorithm, we implemented elitism by storing the best solution of the CHNRS in a separate array called “Best”. Throughout the selection, crossover, and mutation processes in each generation, elitist chromosomes remained unaltered. At the end of the evolutionary process within a generation, the elitist “Best” was replaced with the best solutions obtained from the current population. This integration of elitism ensured that the most favorable solutions persisted and contributed to the overall performance and convergence of the proposed algorithm.

4.2.7. LS procedure

Previous studies have demonstrated that incorporating a local search into evolutionary optimization algorithms can enhance their exploitation capabilities [62]. By exploring promising solution spaces

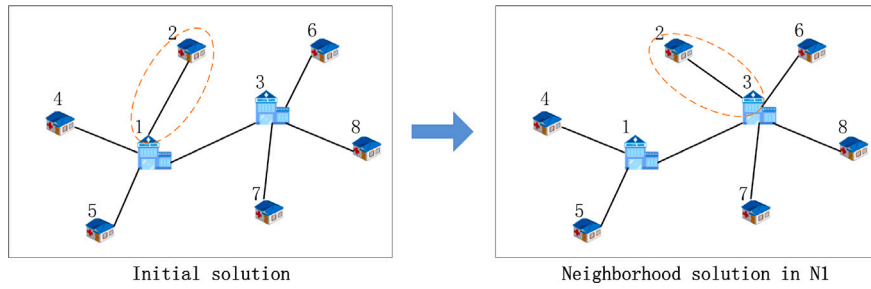


Fig. 9. An example of neighborhood N1.

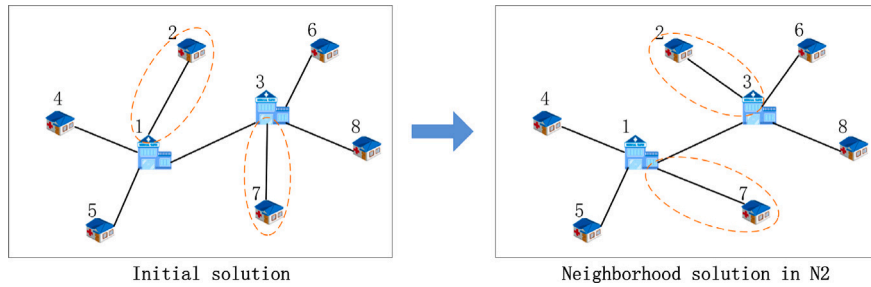


Fig. 10. An example of neighborhood N2.

through neighborhood actions, the hybridization with local search strategies enables deep search abilities. Therefore, in our algorithm, we employed a local search procedure to accelerate the hospital assignment process.

Our LS procedure involved two neighborhoods, N1 and N2, characterized by shift and exchange moves, respectively. N1 is generated through shift moves, whereby the assignment of a non-hub node changes from one hub to another within the same interval. This move is permissible only when there are at least two hubs within the interval. Fig. 9 illustrates an example of a shift move, where the assignment of node 2 is switched from hub 1 to hub 3. On the other hand, N2 operates through exchange moves that swap the assignments of two non-hub nodes connected to different hubs. During the initial assignment as shown in Fig. 10, node 2 is assigned to hub 1, and node 7 is assigned to hub 3. After the exchange move, the new assignments were node 2 to hub 3 and node 7 to hub 1.

The LS procedure was applied to every individual in the initial population or to those generated during the crossover and mutation process. To mitigate premature convergence, we applied shift and exchange moves based on specific probabilities. Additionally, we introduced the maximum number of neighborhoods, Loc_n , as the termination condition. This ensured that we explored a maximum number of Loc_n neighborhood solutions for each individual. If we encountered a solution superior to the current solution within the Loc_n count, we terminated the LS. If the LS process yielded a solution that was worse than the initial solution, we disregarded it and retained the initial solution. Algorithm 2 presents a detailed description of the LS procedure.

4.3. Computational complexity

The computational complexity indicates the operational efficiency of an algorithm and is an important factor in determining the algorithm's merit [63]. For genetic algorithms, the computational complexity is driven by the number of calls to genetic operators. Thus, the computational complexity of HGA primarily stems from four elements: initial population, fitness function evaluation, crossover operator, and mutation [64]. Assume that the individual dimension of HGA is n , the population size is N , and the maximum iteration number is Gen . During the initialization stage, the time required to set each initial parameter

is constant. Assuming that the time for initializing the dimension of an individual is n_1 , the complexity of this stage is $O(N \times n \times n_1) = O(N \times n)$. During crossover process, only one type of crossover operator is randomly selected for execution, so the computational complexity of crossover is $O(Gen \times N \times n)$. Similarly, the complexity of mutation is $O(Gen \times N \times n)$. Meanwhile, the objective function evaluation $f(n)$ is linear and its complexity is $O(Gen \times N \times n)$, as it evaluates the objective value of each individual in the population per generation.

Since the LS procedure is applied to each individual generated during initialization, crossover, and mutation processes, the complexity of the HGA is additionally increased. Assume that the time required to perform a shift move is T_{N1} and an exchange move is T_{N2} . Both T_{N1} and T_{N2} are constants. Since the probability of executing both shift and exchange moves is not greater than 1, the maximum complexity of the LS procedure during the initialization is $O(N \times Loc_n \times (T_{N1} + T_{N2})) = O(N \times Loc_n)$, where Loc_n denotes the maximum number of neighborhoods explored by the LS procedure each time. In addition, the maximum complexity due to the embedded LS procedure in both crossover and mutation is $O(Gen \times N \times Loc_n \times (T_{N1} + T_{N2})) = O(Gen \times N \times Loc_n)$. Therefore, the computational complexity of HGA is $O(N \times n) + 3 \times O(Gen \times N \times n) + O(N \times Loc_n) + 2 \times O(Gen \times N \times Loc_n)$. Since $Loc_n \leq n$, the computational complexity of HGA is less than $O(N \times n) + 3 \times O(Gen \times N \times n) + O(N \times n) + 2 \times O(Gen \times N \times n) = 2 \times O(N \times n) + 5 \times O(Gen \times N \times n)$.

5. Numerical experiments

In this section, a series of computational experiments were conducted to showcase the effectiveness of the proposed models and algorithm. We used in our experiments a personal computer with an Intel(R) Core(TM) i7-8700 CPU @ 3.20 GHz and 8 GB of RAM running on a Microsoft Windows 10 operating system. All algorithms were coded in C++ using Visual Studio 2019, and the results obtained by CPLEX 12.8.0 were used for comparison.

5.1. Study area and data collection

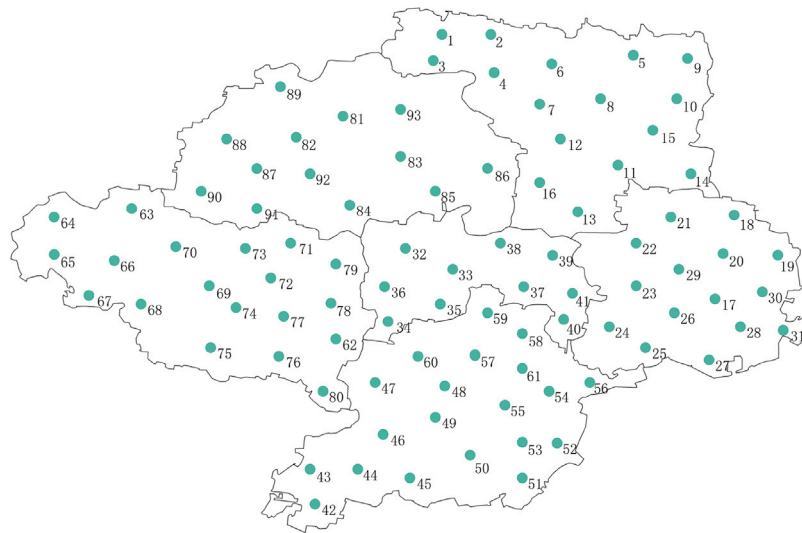
In this study, we used a dataset comprising 93 hospitals, taken from six counties in Hebei, China. Each county was considered to be

Algorithm 2 The LS pseudocode.**Require:** A Chromosome n **Ensure:** An improved chromosome n with better assignment

```

1:  $f_{best} \leftarrow f(n)$   $\triangleright f(n)$  is the objective value of Chromosome  $n$ 
2:  $Temp \leftarrow Chromosome\ n$ ;
3:  $t \leftarrow 1$ ;
4: while  $t \leq Loc\_n$  do  $\triangleright Loc\_n$  is the maximum number of neighbors
5:   Randomly select a cluster  $k$  in Chromosome  $n$ ;
6:   if the number of hubs in cluster  $k$  is more than 1 then
7:     if  $rand(0, 1) < loc\_1$  then  $\triangleright loc\_1$  represents the probability of applying the first local search strategy
8:       Select a non-hub node  $i$  in cluster  $k$  of Chromosome  $n$ ;
9:       Assign  $i$  to another hub node  $j$  in cluster  $k$  and get an updated Chromosome  $n'$ ;
10:      Chromosome  $n \leftarrow Chromosome\ n'$ ;
11:    end if
12:    if  $rand(0, 1) < loc\_2$  then  $\triangleright loc\_2$  represents the probability of applying the second local search strategy
13:      Select two non-hub nodes in cluster  $k$  of Chromosome  $n$  that are assigned to different
14:      hub nodes;
15:      Exchange the assignment of these two nodes and get an updated Chromosome  $n'$ ;
16:      Chromosome  $n \leftarrow Chromosome\ n'$ ;
17:    end if
18:     $f_{best1} \leftarrow f(n)$ ;
19:    if  $f_{best1} > f_{best}$  then
20:      Chromosome  $n \leftarrow Temp$ ;
21:    else
22:       $f_{best} \leftarrow f_{best1}$ ;
23:       $Temp \leftarrow Chromosome\ n$ ;
24:      break;
25:    end if
26:  end if
27: end while

```

**Fig. 11.** Locations of 93 hospitals in 6 counties.

a cluster with numerous PHs and CHs. Fig. 11 depicts the geographical distribution of these hospitals and the partitioning of the six clusters that formed the network. For the purposes of this study, we referred to this network as a Large Scale Network (LSN) and generated a series of small instances based on the LSN.

The data for the CHNRS composed four main parts: location, request, resource, and cost. The location data provided the physical location of all hospitals in the network; however, in our study, we used the distance between hospitals as a proxy for location information. The requested data pertained to the number of patients with various healthcare needs. Resource data specify the maximum service capacity of each hospital. The cost data included construction, connection, and patient treatment costs. To obtain the data mentioned above, we first used Google Maps API's ranging function to extract the actual distance

between each pair of nodes in the LSN, and then converted these distances into connection costs using an average rate of 10,000 CNY per kilometer. Based on the ambulance fee rate for 2021, the referral fee was set to 3 CNY per kilometer. To ensure that our request and resource data accurately reflected real-world conditions, we surveyed the average number of daily visits and the service capacity of each hospital. Using this information as reference, we estimated the approximate demand and capacity of each hospital. Moreover, by scaling the obtained demand or service capacity data up or down, we generated multiple sets of demand and capacity at different levels.

After obtaining the LSN dataset (LSND), we created a small-scale dataset involving 20 hospitals. These hospitals were selected from the first and second clusters of the LSN, and relevant data were extracted directly from the LSND. The experiments based on this small dataset

Table 4
Clustering rules for the 20 nodes data set.

P_s	NoC		
	2	3	4
12	{1-6}	{1-4}	{1-3}
	{7-12}	{4-8}	{4-6}
		{9-12}	{7-9}
16			{10-12}
	{1-8}	{1-5}	{1-4}
	{9-16}	{6-10}	{5-8}
		{11-16}	{9-12}
20			{13-16}
	{1-10}	{1-7}	{1-5}
	{11-20}	{8-14}	{6-10}
		{15-20}	{11-15}
			{16-20}

were organized into three problem sizes, where $N \in \{12, 16, 20\}$. For each problem size, we considered different numbers of clusters, ranging from 2 to 4. The specific clustering rules for this dataset are presented in Table 4, where “P_s” denotes the problem size, “NoC” represents the amount of clusters, and “C_r” is the abbreviation of the clustering rule.

5.2. Parameter settings

Our HGA incorporated 11 primary parameters: p_c denotes the probability of crossover operators; c_1 and c_2 represent the selection probabilities for one-point and two-point crossover operators, respectively; p_m is the probability of mutation operators; m_1 and m_2 denote the selection probabilities for two-exchange and site mutation operators; loc_1 and loc_2 correspond to the probabilities of two types of local search strategies; N denotes the population size; and Gen is the number of generations.

Appropriate parameter setting is always a prerequisite for achieving good computational performance. The values of the HGA parameters were determined after a series of preliminary tuning experiments carried out on the 20 nodes datasets. As the complexity of the CHNRS problem increases with its size, we indexed the population size to the problem size, i.e., we used small populations for small-scale problems and larger populations for larger ones. Given that the algorithm converges faster for small-scale problems, we employed fewer iterations for small-scale problem instances and more iterations for large-scale problem instances. According to the literature [65–67], the mutation and crossover parameters of GA are usually chosen in the following sets: $p_m \in \{0.3, 0.4, 0.5, 0.6\}$ and $p_c \in \{0.4, 0.5, 0.6, 0.7\}$. Additionally, we considered several parameter settings for other parameters: $c_1 \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$, $m_1 \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$, $loc_1 \in \{0.4, 0.5, 0.6, 0.7\}$, $loc_2 \in \{0.4, 0.5, 0.6, 0.7\}$. Notably, the algorithm executes only one type of crossover, mutation operators during each iteration. Therefore, the values of parameters c_2 and m_2 were determined based on the relationships $m_1 + m_2 = 1$ and $c_1 + c_2 = 1$. The various combinations of these parameter values mentioned above constitute the candidate configurations. The efficacy of each candidate configuration is evaluated by running the same number of runs on training instances, with the one yielding the best performance considered the optimal parameter setting. Table 5 lists the values of all parameters.

5.3. Computational results

5.3.1. HGA performance

To comprehensively verify the performance of the HGA in solving the CHNRS problem, we first conducted a set of numerical experiments using the small-scale dataset with $N \in \{12, 16, 20\}$. For each problem scale, multiple tests were performed by varying the demand across

three levels: d_1 , d_2 , and d_3 , representing low, medium, and high demand levels, respectively. Correspondingly, each demand level was paired with three capacity levels denoted as q_1 , q_2 , and q_3 , indicating the low, medium, and high capacity levels, respectively.

Table 6 collects the results from the aforementioned small-scale experiments. In the table, “n” denotes the number of nodes in the network, and “C” represents the number of clusters. The demand levels of patients are represented by “D”, where “d1”, “d2” and “d3” correspond to three distinct levels. Similarly, “Q” denotes the maximum service capacity of hospitals, with “q1”, “q2” and “q3” corresponding to three different capacity levels. The “Obj” column displays the objective values and the solving time is shown in the “CPU(s)” column. The “Gap” column indicates the difference between the objective values of HGA and CPLEX; the “Δ” column shows the difference in solving times, where “.” signifies that HGA solved the problem in less time compared to CPLEX. Notably, all the results of HGA were obtained by solving the NCM and NFM under the algorithm framework, while all results of the CPLEX solver were obtained by solving the MILP model. The results provide strong evidence that our proposed algorithm is capable of finding optimal solutions for each small-scale problem instance, as using the CPLEX solver. Furthermore, our algorithm demonstrates excellent efficiency in solving time, achieving approximately 50% time savings compared to the CPLEX solver in finding optimal solutions.

To further highlight the unique advantages of the HGA, we carried out a series of comparative experiments. First, the proposed HGA was compared with the well-established Particle Swarm Optimization (PSO) algorithm, demonstrating that HGA was competitive among evolutionary algorithms. As a swarm intelligence algorithm, PSO is inspired by the social behaviors observed in bird flocking and fish schooling. This biologically inspired mechanism has led to PSO’s widespread application across various domains, as highlighted in recent studies [68]. Its effectiveness in tackling complex optimization problems has been well-established. In addition, due to its simplicity, speed, and efficiency, PSO is often used as a benchmark evolutionary algorithm for comparison with GA, which is one of the primary reasons we have chosen it for our study. The design of the PSO in this paper is mainly based on [69]. Second, the comparison between HGA and GA was included to evaluate the contribution of the local search procedure integrated into HGA. At last, the results obtained by solving the MILP with the CPLEX solver are used to examine the solution accuracy of other algorithms.¹ Similar to the HGA, both PSO and GA deal with the two submodels, NCM and NFM, with NFM being solved by calling the CPLEX solver to return the objective value. Both small- and large-scale problem instances were selected to complete these tests, and the best, worst, and standard deviation values were derived by conducting 10 runs for each instance and each algorithm. The statistics in Table 7 show that the best value obtained by the HGA was the same as the exact value achieved by the CPLEX solver. In addition, the accuracy and stability of the HGA were significantly better than the other two comparison algorithms across all test instances. Specifically, in the cases of $n = 12$ and $n = 16$, the standard deviation value of the HGA was 0, indicating the remarkable stability of the HGA when addressing small-scale instances. As the problem scale increased, the standard deviation value of the HGA increased, but remained comparatively lower than those of the GA and PSO. Additionally, the comparison between the HGA and GA underscores the efficacy of incorporating local search strategies, showing a marked enhancement in solution accuracy and robustness.

To better study the convergence speed of the HGA, we performed time-to-target tests for both small- and large-scale instances. Such tests illustrate the probability of the time required to achieve a solution as good as the target value. It is useful for characterizing the running times of stochastic algorithms for combinatorial optimization problems.

¹ The CPLEX solver fails to solve the large instance within 24 h.

Table 5
Parameters settings for the HGA.

Parameter	Description	Tested values	Final
p_c	Crossover probability	{0.4, 0.5, 0.6, 0.7}	0.5
c_1	One-point crossover operator probability	{0.3, 0.4, 0.5, 0.6, 0.7}	0.6
c_2	Two-point crossover operator probability	{0.3, 0.4, 0.5, 0.6, 0.7}	0.4
p_m	Mutation probability	{0.3, 0.4, 0.5, 0.6}	0.4
m_1	Two-exchange mutation operator probability	{0.3, 0.4, 0.5, 0.6, 0.7}	0.3
m_2	Site mutation operator probability	{0.3, 0.4, 0.5, 0.6, 0.7}	0.7
loc_1	The first local search strategy probability	{0.4, 0.5, 0.6, 0.7}	0.5
loc_2	The second local search strategy probability	{0.4, 0.5, 0.6, 0.7}	0.5
N^s	Population size for small problems	{20, 30, 40}	30
N^l	Population size for large problems	{100, 120, 140}	120
Gen^s	Iterations for small problems	{200, 300, 400, 500, 600}	400
Gen^l	Iterations for large problems	{600, 800, 1000, 1200, 1400}	1000

Table 6
Results of the algorithm on the 20 nodes dataset.

C	D	Q	n = 12						n = 16						n = 20					
			Obj($\times 10^6$)			CPU (s)			Obj($\times 10^6$)			CPU (s)			Obj($\times 10^6$)			CPU (s)		
			CPLEX	HGA	Gap	CPLEX	HGA	Δ (%)	CPLEX	HGA	Gap	CPLEX	HGA	Δ (%)	CPLEX	HGA	Gap	CPLEX	HGA	Δ (%)
2	d1	q1	231.83	231.83	0	2.22	1.21	-45.72	300.72	300.72	0	3.03	1.76	-41.82	395.70	395.70	0	3.01	0.78	-74.22
		q2	235.68	235.68	0	2.18	0.40	-81.74	303.77	303.77	0	2.21	0.96	-56.70	395.36	395.36	0	1.99	1.71	-14.02
		q3	244.61	244.61	0	1.98	0.49	-75.35	310.58	310.58	0	1.99	0.49	-75.63	397.82	397.82	0	1.99	1.51	-24.27
	d2	q1	473.18	473.18	0	2.75	1.82	-33.67	609.90	609.90	0	2.24	6.13	173.57	808.31	808.31	0	2.74	1.38	-49.64
		q2	462.40	462.40	0	2.63	0.64	-75.70	599.81	599.81	0	2.13	1.64	-23.15	788.98	788.98	0	2.04	0.90	-55.74
		q3	464.11	464.11	0	2.50	0.62	-75.36	602.49	602.49	0	2.35	0.57	-75.87	788.99	788.99	0	1.84	1.57	-14.73
	d3	q1	750.74	750.74	0	4.66	1.17	-74.83	980.95	980.95	0	2.23	1.04	-53.18	1347.23	1347.23	0	2.03	0.47	-76.65
		q2	694.92	694.92	0	2.11	1.21	-42.80	900.48	900.48	0	2.13	1.68	-20.99	1190.41	1190.41	0	2.11	3.02	43.03
		q3	692.88	692.88	0	2.09	0.45	-78.28	898.81	898.81	0	2.00	1.56	-22.15	1182.11	1182.11	0	2.21	1.43	-35.29
3	d1	q1	235.48	235.48	0	2.22	1.52	-31.58	304.41	304.41	0	2.80	2.01	-28.11	398.58	398.58	0	2.43	1.88	-22.76
		q2	245.97	245.97	0	1.63	1.94	19.08	313.92	313.92	0	2.30	1.73	-24.70	402.15	402.15	0	2.68	1.49	-44.44
		q3	258.51	258.51	0	2.22	0.51	-76.98	326.60	326.60	0	1.91	0.71	-63.04	414.19	414.19	0	2.17	1.56	-28.25
	d2	q1	474.66	474.66	0	2.35	2.83	20.43	610.37	610.37	0	2.00	1.74	-13.25	809.02	809.02	0	2.30	10.79	369.00
		q2	469.54	469.54	0	2.15	0.65	-69.95	606.90	606.90	0	2.20	0.64	-70.95	794.30	794.30	0	2.17	0.77	-64.56
		q3	478.24	478.24	0	2.17	0.67	-69.08	615.44	615.44	0	1.95	1.53	-21.54	795.58	795.58	0	2.11	2.00	-5.02
	d3	q1	751.35	751.35	0	2.90	0.38	-87.00	980.95	980.95	0	2.17	0.68	-68.89	1347.23	1347.23	0	2.17	0.50	-77.19
		q2	701.21	701.21	0	2.18	1.34	-38.67	905.96	905.96	0	1.88	1.88	0.11	1190.82	1190.82	0	3.11	2.18	-29.81
		q3	703.47	703.47	0	2.17	0.94	-56.64	909.31	909.31	0	2.02	1.80	-11.04	1189.87	1189.87	0	2.89	2.20	-23.77
4	d1	q1	240.73	240.73	0	2.17	0.93	-56.96	308.67	308.67	0	2.38	0.92	-61.34	402.34	402.34	0	2.09	2.09	0.05
		q2	255.94	255.94	0	2.39	1.00	-58.12	323.23	323.23	0	2.42	0.91	-62.40	414.20	414.20	0	2.23	1.97	-11.61
		q3	272.23	272.23	0	2.18	0.91	-58.26	340.33	340.33	0	2.01	0.79	-60.80	430.80	430.80	0	2.03	1.89	-6.90
	d2	q1	474.54	474.54	0	2.16	1.79	-17.04	615.49	615.49	0	2.06	1.69	-17.77	812.51	812.51	0	2.45	11.86	384.04
		q2	479.55	479.55	0	2.19	1.21	-44.79	615.01	615.01	0	1.95	0.77	-60.77	801.38	801.38	0	2.62	2.09	-20.34
		q3	493.70	493.70	0	2.19	1.06	-51.42	628.03	628.03	0	2.00	0.78	-61.20	811.16	811.16	0	2.03	1.69	-16.55
	d3	q1	751.35	751.35	0	2.40	0.70	-70.67	981.69	981.69	0	2.02	0.91	-55.10	1347.23	1347.23	0	2.45	0.49	-80.16
		q2	710.96	710.96	0	2.21	1.46	-33.76	912.92	912.92	0	1.88	1.86	-1.01	1198.32	1198.32	0	2.05	1.62	-20.78
		q3	718.28	718.28	0	2.18	1.12	-48.85	921.17	921.17	0	1.97	2.38	20.91	1200.26	1200.26	0	2.24	2.08	-7.01

Table 7
Solution results for 4 approaches for solving the CHNRS problem.

n	Algorithm	Best value	Worst value	Standard deviation value
12	CPLEX	4.732×10^8	4.732×10^8	0
	HGA	4.732×10^8	4.732×10^8	0
	GA	4.732×10^8	4.739×10^8	3.320×10^5
	PSO	4.732×10^8	4.740×10^8	2.315×10^5
16	CPLEX	6.099×10^8	6.099×10^8	0
	HGA	6.099×10^8	6.099×10^8	0
	GA	6.107×10^8	6.143×10^8	1.681×10^6
	PSO	6.108×10^8	6.149×10^8	1.348×10^6
20	CPLEX	8.090×10^8	8.108×10^8	0
	HGA	8.090×10^8	8.108×10^8	5.454×10^5
	GA	8.094×10^8	8.114×10^8	7.542×10^5
	PSO	8.110×10^8	8.160×10^8	1.735×10^6
93	CPLEX	—	—	—
	HGA	5.630×10^9	5.655×10^9	2.707×10^6
	GA	5.637×10^9	5.646×10^9	7.405×10^6
	PSO	5.708×10^9	5.780×10^9	2.306×10^7

A detailed explanation of time-to-target plots and the tools used to generate them can be found in [70].

We used the results of the previous tests as a reference to set the target values. The target solution value was set to 8.10×10^8 for the small-scale instance with 20 nodes, and 5.77×10^9 for the large-scale instance with 93 nodes. Each algorithm was run 100 times for each instance, until a solution at least as good as the corresponding target

was found. For each of the 100 runs, the random number generator used in the implementation of the heuristic was initialized with a distinct seed. Therefore, the runs were assumed to be independent. The results are shown in Figs. 12 and 13. The time (in seconds) is represented on the abscissa axis, while the ordinate axis shows the cumulative probability that the algorithm reaches solutions as good as the targets using that time.

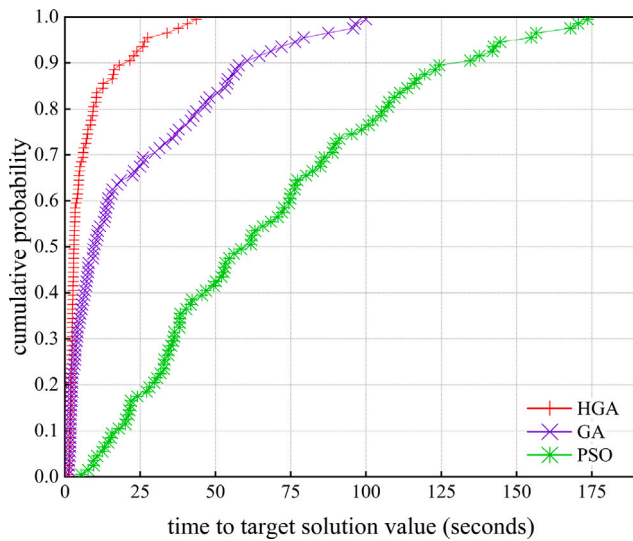


Fig. 12. Time-to-target plot of HGA respect to GA and PSO for an instance with 20 nodes.

Fig. 12 concludes that HGA outperforms GA and PSO, presenting a higher probability of finding the targets in a fixed amount of time. For example, the probability of finding a target value for the 20 node instance within 25 s is 92.5% for HGA, while is around 67.5% and 17.5% for GA and PSO. The probabilities of these three algorithms are about 100%, 83.5%, and 42.5% in at most 50 s, respectively. Additionally, this figure shows that HGA can meet the probability requirements in a much shorter time compared to GA and PSO. For example, the time needed to have 50% of the runs find the solution of the desired quality is close to 3.016 s, 9.891 s, and 58.540 s for the three algorithms, respectively. Similarly, GA is better than PSO in terms of convergence speed. Fig. 13 presents similar conclusions regarding the convergence speed of the algorithms for large-scale instances. The difference is that when the cumulative probability is below 15%, HGA and GA take comparable times. However, when the cumulative probability exceeds 15%, HGA demonstrates a significant advantage over GA. Additionally, when the cumulative probability is below 40%, GA takes more time to reach the target solution compared to PSO. But when the probability exceeds 40%, GA gradually shows a significant advantage over PSO. In summary, these results demonstrate that the LS procedure drastically speeds up the convergence of HGA to good solutions. Meanwhile, the comparison between GA and PSO highlights the advantages of GA over other evolutionary algorithms, which supports the selection of GA for this study.

5.3.2. Efficacy analysis of the proposed model

Fig. 14 shows the result from an instance with 16 nodes characterized by low service capacity and medium demand levels. It is evident that the optimal locations identified for CHs were 3, 9, 14, and 15. Since there was only one CH in the first cluster, all PHs in this cluster were assigned to CH 3. Similarly, all PHs in the second cluster were allocated to the CH 9. In the third cluster, PHs 13 and 16 were assigned to CH 14, and PHs 11–12 were assigned to CH 15. Additionally, we conducted an experiment to assess the efficacy of our algorithm in addressing significant challenges encountered in large-scale problems. Fig. 15 shows the optimal network structure obtained by HGA on the 93 node dataset. We can conclude that the optimal locations for CHs are 8, 15, 26, 40, 41, 48, 52, 66, 70, 73, 83, and 92, and the best allocation is assigning the following: 1–9 and 12 to CH 8; 10, 11 and 13–14 to CH 15; 17–25 and 27–31 to CH 26; 34–37 to CH 40; 32–33 and 38–39 to CH 40; 42–47, 55 and 59–60 to 48; 51, 53–54, 56–58 and 61 to CH 52; 63–65 and 67–68 to 66; 69, 71 and 74 to 70; 71–72, 75–78 and 80

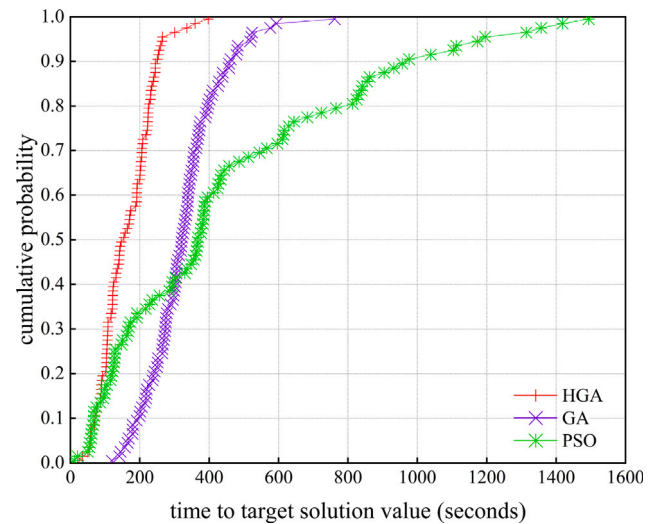


Fig. 13. Time-to-target plot of HGA respect to GA and PSO for an instance with 93 nodes.

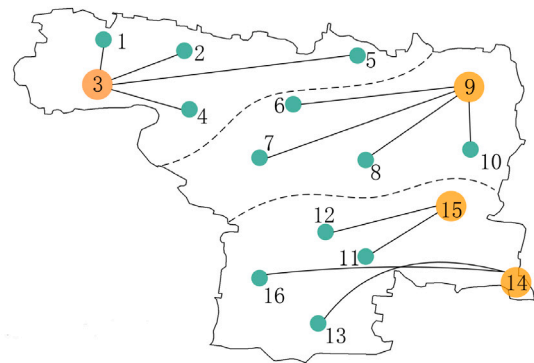


Fig. 14. The healthcare network structure of 16 nodes in 3 counties.

to 73; 81, 84–86, 89 and 93 to 83; and 82, 87, 88, 90 and 91 to 92. In conclusion, our proposed model was capable of constructing a rational healthcare network efficiently, especially for large-scale instances.

5.3.3. Analysis for healthcare network capacity planning

Table 6 provides useful insights for healthcare network management. We observed that the total cost (TC) of the network increased as patient demands grew, regardless of service capacity or clustering rules. However, the amount of cost increase caused by demand expansion varies depending on the service capacity constraint. Generally, the cost increase decreases as service capacity increases. This indicates that a healthcare network with high service capacity is more adaptive and robust to changes in demand.

Moreover, we observed that TC exhibited different trends as the service capacity increased with different demand levels. Specifically, we identified three types of trends: continuous rise, decrease and then rise, and continuous decrease. When the demand level is low, a low service capacity is sufficient to meet patients' needs, and any additional capacity increase would result in a waste of healthcare resources and increased construction costs, resulting in a continuous increase in costs. For medium demand levels, low service capacity limits access to optimal options, causing the total costs to initially decrease as capacity increases. However, if the service capacity is too high, excess capacity will increase investment without practical significance, leading to increased costs. The last trend occurs when none of the three service capacity levels optimally meet patient demands. In this case,

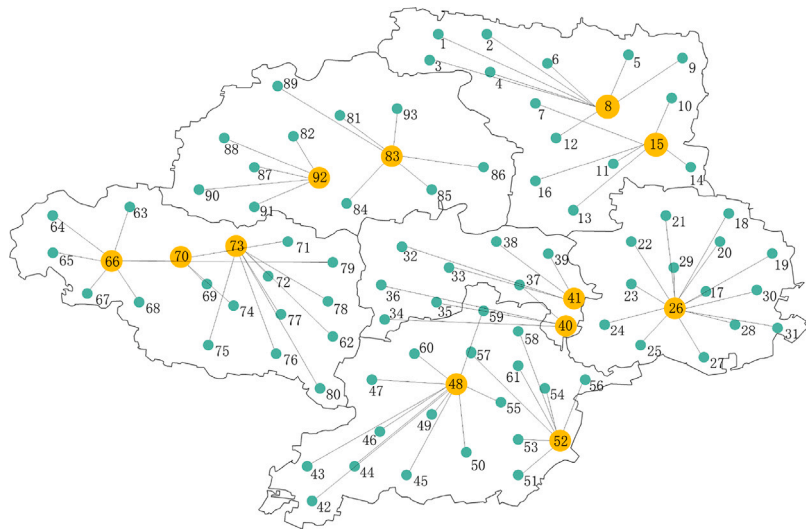


Fig. 15. The healthcare network structure of 93 nodes in 6 counties.

Table 8
Comparative results between the single-hub and multiple-hub forms.

D	L	Obj($\times 10^9$)					
		n = 12		n = 16		n = 20	
		p = 1	p $\geq$ 1	p = 1	p $\geq$ 1	p = 1	p $\geq$ 1
d1	5	1.359	1.359	1.772	1.772	2.493	2.481
	10	2.685	2.685	3.531	3.531	4.972	4.935
	15	4.341	4.330	5.291	5.228	7.451	7.388
d2	5	1.685	1.684	2.045	2.045	2.883	2.860
	10	3.357	3.349	4.079	4.072	5.751	5.691
	15	5.028	5.012	6.112	6.097	8.619	8.522
d3	5	1.917	1.912	2.322	2.317	3.281	3.243
	10	3.819	3.804	4.631	4.614	6.548	6.456
	15	5.722	5.695	6.941	6.911	9.815	9.669

the network becomes more resilient as the hospital capacity expands, giving patients more choices and significantly reducing the total patient treatment costs. Overall, our findings suggest that healthcare networks should adjust their service capacity based on demand levels to ensure optimal resource utilization and cost-efficiency.

5.3.4. Analysis for healthcare network hub forms

We conducted two sets of experiments to compare the single- and multi-hub forms of the network. For each hub scenario, we varied the demand and capacity levels and time periods to assess the network's performance. The computational results are reported in Tables 8 and 9. In these tables, "L" represents the time periods of the network, which reflects our focus on short-term or long-term effects. "p = 1" and "p $\geq$ 1" indicate a single-hub and at least one hub in each cluster, respectively. The other symbols have the same notations as in Table 6.

Table 8 shows the results for three problem sizes in both hub scenarios. Three demand levels and time periods were considered in these experiments. Overall, for a given problem size and time period, the TC obtained from the multi-hub scenario was not higher than that in the single-hub scenario. In general, the results obtained in the two hub scenarios were the same when demand was low and the time period was short. This is because, at lower demand levels, one CH per cluster is sufficient to meet the needs of all patients. In this case, building additional CHs leads to high construction costs with little reduction in patient treatment costs. Moreover, in a short time period, construction costs comprise a significant portion of the TC relative to patient treatment costs. Therefore, the reduction in patient treatment costs hardly covers the increase in construction costs if more than

one CH is built. As demand increases and the time period lengthens, patient treatments cost gradually exceed the fixed construction costs and take a dominant position. In this case, an appropriate increase in the number of CHs can offer patients better options and eventually keep the TC of the CHNRS network at a lower level. In summary, the multi-hub scenario consistently provides the optimal solution, regardless of demand level or time period.

Table 9 considers three service capacity levels and time periods. The experimental results provide insights similar to those in Table 8. Specifically, when the service capacity was high and the time period short, the single-hub and multi-hub scenarios yielded the same solution. However, as the service capacity decreased and the time period increased, the multi-hub scenario demonstrated clear advantages over the single-hub scenario.

5.3.5. Analysis of healthcare referral strategies

A series of experiments were conducted to compare diverse referral strategies with the aim of validating the efficacy of the proposed hybrid referral system. This system considers two-way referrals between hierarchical hospitals and referrals at the same level. Two-way, upward, and downward referral strategies were evaluated across various scenarios for comparison. Table 10 displays the results of the four types of referral strategies, with "LC" representing the cost growth rate of specific referral strategies relative to the hybrid referral strategy. The findings revealed that the hybrid referral strategy was the optimal choice for minimizing the overall system cost, followed by the two-way, upward, and downward referral strategies. Furthermore, the advantage of the hybrid referral strategy became more prominent as the capacity

Table 9
Comparative results between the single-hub and multiple-hub forms.

Q	L	Obj($\times 10^9$)					
		n = 12		n = 16		n = 20	
		p = 1	p $\geq$ 1	p = 1	p $\geq$ 1	p = 1	p $\geq$ 1
q1	5	4.153	4.133	5.374	5.348	7.050	6.988
	10	8.284	8.230	10.724	10.656	14.069	13.914
	15	12.415	12.327	16.074	15.962	21.088	20.840
q2	5	4.104	4.102	5.340	5.334	6.990	6.962
	10	8.176	8.160	10.652	10.627	13.943	13.861
	15	12.248	12.218	15.965	15.918	20.896	20.760
q3	5	4.074	4.074	4.687	4.687	6.948	6.942
	10	8.110	8.110	9.342	9.342	13.852	13.826
	15	12.146	12.146	13.997	13.997	20.756	20.707

Table 10
Comparative results among different referral strategies.

n	Q	Hybrid referral	Two-way referral	I_C	Upward referral	I_C	Downward referral	I_C
12	q1	4.747×10^8	4.782×10^8	0.753%	4.796×10^8	1.036%	5.138×10^8	8.254%
	q2	4.695×10^8	4.692×10^8	−0.069%	4.710×10^8	0.315%	4.823×10^8	2.719%
	q3	4.782×10^8	4.802×10^8	0.417%	4.805×10^8	0.480%	4.927×10^8	3.031%
16	q1	6.104×10^8	6.153×10^8	0.815%	6.217×10^8	1.854%	6.637×10^8	8.749%
	q2	6.069×10^8	6.070×10^8	0.016%	6.089×10^8	0.335%	6.234×10^8	2.719%
	q3	6.154×10^8	6.156×10^8	0.02%	6.158×10^8	0.053%	6.342×10^8	3.041%
20	q1	8.090×10^8	8.124×10^8	0.416%	8.244×10^8	1.901%	8.741×10^8	8.044%
	q2	7.943×10^8	7.945×10^8	0.019%	7.988×10^8	0.562%	8.150×10^8	2.609%
	q3	7.956×10^8	7.960×10^8	0.057%	7.991×10^8	0.438%	8.260×10^8	3.824%

setting of the healthcare system decreased. For instance, in the case where $n = 12$ and $Q = q1$, the downward referral strategy resulted in an 8.254% increase in system costs compared with the hybrid referral strategy, while at $Q = q3$, this percentage increase was 3.031%. This may be attributed to the fact that the hybrid referral strategy offers patients more referral choices, enabling them to select the most efficient and cost-effective medical options, thereby facilitating the efficient utilization of medical resources within the healthcare system.

6. Managerial insights

This section provides managerial insights based on our model and findings that could be helpful for managers who make decisions about the underlying healthcare network for implementing CMAs.

1. We introduce, for the first time, a CHNRS framework that addresses the hierarchical healthcare network design, healthcare districting, and patient referral problems simultaneously. Our model provides managers with an opportunity to understand how their decisions affect the establishment and operating costs of healthcare systems. Therefore, they can make informed decisions about constructing an effective cluster-based healthcare network and rational patient flow to reduce costs and improve efficiency. Additionally, our model offers significant benefits to government policymakers by providing an effective tool for implementing CMAs across broader geographical areas in China.
2. Our computational results show capacity settings play a crucial role in hospital location allocation and patient referral decisions. Specifically, the study reveals that a healthcare system equipped with a more relaxed service capacity exhibits enhanced adaptability and resilience in response to external changes. By allowing flexibility and accommodating fluctuations in demand, such healthcare systems can respond better to variations in patient needs, resource availability, and other dynamic factors. Simultaneously, implementing relaxed service capacity settings gives patients more choice in selecting hospitals. This promotes the efficient utilization of medical resources and contributes to an improved overall healthcare quality.
3. In the long term, relaxed capacity settings can enhance decision-making flexibility and patient choice within a healthcare network. However, a higher capacity setting is likely to result in wasting of medical resources, increasing construction costs and maintaining higher overall network expenses in the short term. Therefore, decision-makers should carefully design capacity settings based on their long-term goals or short-term preferences, balancing improvements in healthcare quality and operational efficiency with reduced operational costs. If decision-makers prioritize a system's long-term sustainable development and healthcare quality, they may choose a higher-capability setting. Conversely, focusing on short-term cost management may favor relatively lower-capacity settings.
4. The comparative results between single-hub and multi-hub configurations in hierarchical healthcare networks reveal the advantages of the latter. Regardless of the level of demand and the length of the time period, a multi-hub scenario always provides us with an optimal solution. Increasing the number of CHs in each cluster ensures that patients have access to diverse healthcare services, reducing costs and improving their satisfaction. Moreover, the presence of multiple CHs mitigates the risk of overburdening a single CH with excessive demand, thereby enhancing the overall resilience and responsiveness of the healthcare system. This suggests that decision-makers, especially in situations facing high demand and longer planning periods, should consider creating multiple CHs in each cluster when designing and implementing CMAs.
5. Optimizing referral strategies is essential to improve the efficiency and cost-effectiveness of healthcare systems. The proposed hybrid referral strategy allows for two-way referrals between hospitals at different levels and among those at the same level, providing patients with diverse healthcare options. This approach enhances patient engagement and satisfaction while promoting the equitable allocation of healthcare resources. Consequently, decision-makers should prioritize the patient's central role and construct flexible referral mechanisms to guide patients toward more rational and cost-effective medical operations. Moreover, in cases of

limited healthcare capacity or overwhelming demand, the hybrid referral strategy helps optimize the utilization of healthcare resources by broadening patient referral pathways. Consequently, healthcare managers facing resource constraints or emergencies should consider reinforcing the implementation of hybrid referral strategies to alleviate system pressure and ensure timely and effective healthcare delivery to patients.

7. Conclusions

This paper studied the CHNRS problem that arises when implementing CMAs in China to improve healthcare services, especially in rural areas. Specifically, we simultaneously considered healthcare districting and patient referrals in the healthcare network design problem. In this problem, we focused on the location and allocation decisions of hospitals as well as the referral decisions of patients. Correspondingly, we developed an MILP model to minimize the network construction costs and patients' total treatment costs. Given the solving complexity of the MILP model, we decomposed it into the NCM and NFM. Also, we improve the NFM by reducing the number of variables by approximately 90%. To solve these models, we developed an algorithm framework in which the NCM was solved using the HGA, and the NFM was solved using an LP solver. Finally, we performed a set of experiments to verify the rationality of the models and effectiveness of the algorithm. The results show that our models contribute to the efficient design of healthcare networks. Moreover, the proposed heuristic framework outperforms the CPLEX solver, GA and PSO, demonstrating significant superiority in terms of convergence speed and robustness.

Our study also opens up several avenues for future research. First, our initial focus on minimizing total costs may not have adequately reflected the comprehensive and high-quality development needs of healthcare systems. One critical limitation of our current model is the need to balance more metrics, such as system efficiency, accessibility, equity, and patient satisfaction, in the objective. Future studies should aim to expand the model into a multi-objective optimization form by incorporating a more comprehensive objective function. Accordingly, multi-objective solution methods can be applied to solve the proposed model. Second, the parameters used in the our model were assumed to be known and fixed. However, in reality, some critical parameters, such as demand, are heavily affected by the dynamic environment. Therefore, precise information regarding these parameters is often difficult to obtain and anticipate. In future work, uncertainties could be considered as a fundamental element in the healthcare network design problem and added to the model. Uncertain programming methods, such as possibility, fuzzy, and robust methods, can be applied to develop a robust optimization model.

CRedit authorship contribution statement

Luqi Wang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Guoqing Yang:** Writing – review & editing, Writing – original draft, Validation, Methodology, Funding acquisition. **Jianmin Xu:** Writing – review & editing, Writing – original draft, Supervision, Investigation.

Acknowledgments

This work are supported by the National Natural Science Foundation of China (Grant No. 71801077), the Top-notch talents of Hebei province (Grant No. 702800118009), Natural Science Foundation of Hebei Province (Grant No. G2022201002) and the High-Level Innovative Talent Foundation of Hebei University (Grant No. 521000981073).

Data availability

Data will be made available on request.

References

- [1] UN. Sustainable development goals. 2020, Available online: <https://www.un.org/sustainabledevelopment/>. [Accessed 14 2021].
- [2] Statistical communiqué of the People's Republic of China on the 2022 national economic and social development. National Bureau of Statistics of China; 2023, Available online: http://www.stats.gov.cn/english/PressRelease/202302/t20230227_1918979.html.
- [3] China to set up more medical alliances in cities, counties. National Health Commission of the People's Republic of China; 2019, Available online: <http://en.nhc.gov.cn/2019-06/11/c75347.htm>.
- [4] Darmian SM, Fattahi M, Keyvanshokoh E. An optimization-based approach for the healthcare districting under uncertainty. *Comput Oper Res* 2021;135:105425.
- [5] Steiner MTA, Datta D, Neto PJS, Scarpin CT, Figueira JR. Multi-objective optimization in partitioning the healthcare system of Parana state in Brazil. *Omega* 2015;52:53–64.
- [6] Kalcics J, Ríos-Mercado RZ. Districting problems. *Locat Sci* 2019;70:5–743.
- [7] Ríos-Mercado RZ. Optimal districting and territory design (vol. 284). Springer Nature; 2020.
- [8] Sung CS, Jin HW. Dual-based approach for a hub network design problem under non-restrictive policy. *European J Oper Res* 2001;132(1):88–105.
- [9] Yu J, Liu Y, Chang GL, Yang XG. Cluster-based optimization of urban transit hub locations: Methodology and case study in China. *Transp Res Rec* 2008;2042(1):109–16.
- [10] Shang X, Yang K, Jia B, Gao Z. Distributionally robust cluster-based hierarchical hub location problem for the integration of urban and rural public transport system. *Comput Ind Eng* 2021;155:107181.
- [11] Yu M, Zhou W, Jiang B. Referral strategies and capacity decisions in a tiered hospital system with gatekeeping designs—Exemplified with Chinese healthcare system. *Comput Ind Eng* 2022;171:108447.
- [12] Zhou C, Lan Y, Li W, Zhao R. Medicare policies in a two-tier healthcare system with overtreatment. *Omega* 2022;109:102607.
- [13] Li N, Kong N, Li Q, Jiang Z. Evaluation of reverse referral partnership in a tiered hospital system—a queuing-based approach. *Int J Prod Res* 2017;55(19):5647–63.
- [14] Liu X, Cai X, Zhao R, Lan Y. Mutual referral policy for coordinating health care systems of different scales. *Int J Prod Res* 2015;53(24):7411–33.
- [15] Zarrinpoor N, Fallahnezhad MS, Pishvae MS. The design of a reliable and robust hierarchical health service network using an accelerated benders decomposition algorithm. *European J Oper Res* 2018;265(3):1013–32.
- [16] Rahman SU, Smith DK. Use of location-allocation models in health service development planning in developing nations. *European J Oper Res* 2000;123(3):437–52.
- [17] Daskin MS, Dean LK. Location of health care facilities. In: Sainford F, Brandeau M, P W, editors. *Handbook of OR/MS in health care: a handbook of methods and applications*, 2004, p. 43–76.
- [18] Ahmadi-Javid A, Seyedi P, Syam SS. A survey of healthcare facility location. *Comput Oper Res* 2017;79:223–63.
- [19] Galvão RD, Nobre FF, Vasconcellos MM. Modelos matemáticos de localização aplicados à organização espacial de unidades de saúde. *Rev Saude Publica* 1999;33:422–34.
- [20] Zarrinpoor N, Fallahnezhad MS, Pishvae MS. Design of a reliable hierarchical location-allocation model under disruptions for health service networks: A two-stage robust approach. *Comput Ind Eng* 2017;109:130–50.
- [21] Galvão RD, Espejo LGA, Boffey B. A hierarchical model for the location of perinatal facilities in the municipality of Rio de Janeiro. *European J Oper Res* 2002;138(3):495–517.
- [22] Galvão RD, Espejo LGA, Boffey B, Yates D. Load balancing and capacity constraints in a hierarchical location model. *European J Oper Res* 2006a;172(2):631–46.
- [23] Mestre AM, Oliveira MD, Barbosa-Póvoa AP. Location-allocation approaches for hospital network planning under uncertainty. *European J Oper Res* 2015;240(3):791–806.
- [24] Zhang H, Zhang K, Chen Y, Ma L. Multi-objective two-level medical facility location problem and tabu search algorithm. *Inform Sci* 2022;608:734–56.
- [25] Farahani RZ, Hekmatfar M, Fahimnia B, Kazemzadeh N. Hierarchical facility location problem: Models, classifications, techniques, and applications. *Comput Ind Eng* 2014;68:104–17.
- [26] Ortiz-Astorquiza C, Contreras I, Laporte G. Multi-level facility location problems. *European J Oper Res* 2018;267(3):791–805.
- [27] Karatas M, Dasci A. A two-level facility location and sizing problem for maximal coverage. *Comput Ind Eng* 2020;139:106204.
- [28] Pourrezaie-Khaligh P, Bozorgi-Amiri A, Yousefi-Babadi A, Moon I. Fix-and-optimize approach for a healthcare facility location/network design problem considering equity and accessibility: A case study. *Appl Math Model* 2022;102:243–67.
- [29] Farughi H, Tavana M, Mostafayi S, Santos Arteaga FJ. A novel optimization model for designing compact, balanced, and contiguous healthcare districts. *J Oper Res Soc* 2020;71(11):1740–59.
- [30] Yanik S, Bozkaya B. A review of districting problems in health care. *Optim Dist Territ Des* 2020;3:1–55.

- [31] Harper PR, Shahani AK, Gallagher JE, Bowie C. Planning health services with explicit geographical considerations: a stochastic location-allocation approach. *Omega* 2005;33(2):141–52.
- [32] Benzarti E, Sahin E, Dallery Y. Operations management applied to home care services: Analysis of the districting problem. *Decis Support Syst* 2013;55(2):587–98.
- [33] Datta D, Figueira JR, Gourtani AM, Morton A. Optimal administrative geographies: an algorithmic approach. *Sociol- Econ Plan Sci* 2013;47(3):247–57.
- [34] Hao Y, Liu C, Zhao L, Liu W. A dual-clustering algorithm for a robust medical grid partition problem considering patient referral. *Sociol- Econ Plan Sci* 2023;88:101675.
- [35] Shumsky RA, Pinker EJ. Gatekeepers and referrals in services. *Manag Sci* 2003;49(7):839–56.
- [36] Qiu Y, Song J, Liu Z. A simulation optimisation on the hierarchical health care delivery system patient flow based on multi-fidelity models. *Int J Prod Res* 2016;54(21):6478–93.
- [37] Zhou Z, Zhao Y, Shen C, Lai S, Nawaz R, Gao J. Evaluating the effect of hierarchical medical system on health seeking behavior: a difference-in-differences analysis in China. *Soc Sci Med* 2021;268:113372.
- [38] Boffey B, Yates D, Galvão RD. An algorithm to locate perinatal facilities in the municipality of Rio de Janeiro. *J Oper Res Soc* 2003;54(1):21–31.
- [39] Galvão RD, Espejo LGA, Boffey B. Practical aspects associated with location planning for maternal and perinatal assistance in Brazil. *Ann Oper Res* 2006b;143:31–44.
- [40] Guo M, Li B, Wu S, J. Song. Effectiveness of referral incentive policy: exploring using queuing network model with blocking. *IEEE Int Conf Serv Syst Serv Manag* 2011;25–7.
- [41] Wang JJ, Li ZP, Shi JJ. Hospital referral and capacity strategies in the two-tier healthcare systems. *Omega* 2021;100:102229.
- [42] Karakaya S, Meral S. A biobjective hierarchical location-allocation approach for the regionalization of maternal-neonatal care. *Sociol- Econ Plan Sci* 2022;79:101093.
- [43] Huang RL, Wang XK, Hou WH, Chen ZY, Wang YT, Wang JQ. Two-way referral cooperative hospital selection with uncertain information: A two-sided matching decision-making approach. *Comput Ind Eng* 2023;183:109430.
- [44] Hodgson MJ, Jacobsen SK. A hierarchical location-allocation model with travel based on expected referral distances. *Ann Oper Res* 2009;167:271–86.
- [45] Alumur SA, Kara BY, Karasan OE. Multimodal hub location and hub network design. *Omega* 2012;40(6):927–39.
- [46] Kim DG, Kim YD. A branch and bound algorithm for determining locations of long-term care facilities. *European J Oper Res* 2010;206(1):168–77.
- [47] Chouksey A, Agrawal AK, Tanksale AN. A hierarchical capacitated facility location-allocation model for planning maternal healthcare facilities in India. *Comput Ind Eng* 2022;167:107991.
- [48] Kim DG, Kim YD. A Lagrangian heuristic algorithm for a public healthcare facility location problem. *Ann Oper Res* 2013;206:221–40.
- [49] Mohammadi M, Dehbari S, Vahdani B. Design of a bi-objective reliable healthcare network with finite capacity queue under service covering uncertainty. *Transp Res E: Logist Transp Rev* 2014;72:15–41.
- [50] Bidhandi HM, Patrick J, Noghani P, Varshoei P. Capacity planning for a network of community health services. *European J Oper Res* 2019;275(1):266–79.
- [51] Abyazi-Sani R, Ghanbari R. An efficient tabu search for solving the uncapacitated single allocation hub location problem. *Comput Ind Eng* 2016;93:99–109.
- [52] Shi Y, Boudouh T, Grunder O. A hybrid genetic algorithm for a home health care routing problem with time window and fuzzy demand. *Expert Syst Appl* 2017;72:160–76.
- [53] Silva A, Aloise D, Coelho LC, Rocha C. Heuristics for the dynamic facility location problem with modular capacities. *European J Oper Res* 2021;290(2):435–52.
- [54] Kummer AF, de Araújo OC, Buriol LS, Resende MG. A biased random-key genetic algorithm for the home health care problem. *Int Trans Oper Res* 2024;31(3):1859–89.
- [55] Shariff SR, Moin NH, Omar M. Location allocation modeling for healthcare facility planning in Malaysia. *Comput Ind Eng* 2012;62(4):1000–10.
- [56] Syam SS, Côté MJ. A location-allocation model for service providers with application to not-for-profit health care organizations. *Omega* 2010;38(3–4):157–66.
- [57] Li C, Sun G, Deng L, Qiao L, Yang G. A population state evaluation-based improvement framework for differential evolution. *Inform Sci* 2023;629:15–38.
- [58] Cura T. Hybridizing local searching with genetic algorithms for the job sequencing and tool switching problem with non-identical parallel machines. *Expert Syst Appl* 2023;223:119908.
- [59] Takan MA. A novel MIP model and a hybrid genetic algorithm for operation outsourcing in production scheduling with carbon tax policy. *Expert Syst Appl* 2024;251:123983.
- [60] Zhou G, Zhu Z, Luo S. Location optimization of electric vehicle charging stations: Based on cost model and genetic algorithm. *Energy* 2022;247:123437.
- [61] Cotta C, Gallardo JE, Mathieson L, Moscato P. Memetic algorithms: A contemporary introduction. In: *Wiley encyclopedia of electrical and electronics engineering*. 2016, p. 1–15.
- [62] Ishibuchi H, Yoshida T, Murata T. Balance between genetic search and local search in memetic algorithms for multiobjective permutation flowshop scheduling. *IEEE Trans Evol Comput* 2003;7(2):204–23.
- [63] Liu J, Fu Y, Li Y, Sun L, Zhou H. An effective theoretical and experimental analysis method for the improved slime mould algorithm. *Expert Syst Appl* 2024;247:123299.
- [64] Gutiérrez-Aguirre P, Contreras-Bolton C. A multioperator genetic algorithm for the traveling salesman problem with job-times. *Expert Syst Appl* 2024;240:122472.
- [65] Gonçalves JF, Resende MGC. A parallel multi-population biased random-key genetic algorithm for a container loading problem. *Comput Oper Res* 2012;39(2):179–90.
- [66] Carrabs F. A biased random-key genetic algorithm for the set orienteering problem. *European J Oper Res* 2021;292(3):830–54.
- [67] Fathi M, Khakifirooz M, Diabat A, Chen H. An integrated queueing-stochastic optimization hybrid genetic algorithm for a location-inventory supply chain network. *Int J Prod Econ* 2021;237:108139.
- [68] Liu ZG, Ji XH, Yang Y, Cheng HT. Multi-technique diversity-based particle-swarm optimization. *Inform Sci* 2021;577:298–323.
- [69] Pan QK, Tasgetiren MF, Liang YC. A discrete particle swarm optimization algorithm for the no-wait flowshop scheduling problem. *Comput Oper Res* 2008;35(9):2807–39.
- [70] Reyes A, Ribeiro CC. Extending time-to-target plots to multiple instances. *Int Trans Oper Res* 2018;25(5):1515–36.

Luqi Wang received the B.S. degree from the Department of Management, Hebei University, Baoding, China, in 2016. She is currently a Ph.D. student in the Department of Economics at Hebei University. Her main research interests include healthcare management, supply chain network design, network optimization and intelligent algorithms. She has authored articles on those areas in journals such as *Information Sciences*, *IEEE Transactions on Fuzzy Systems*.

Guoqing Yang received the B.S. and M.S. degrees from the Department of Mathematics at Hebei University, Baoding, China. He received the Ph.D. degree in Management Science and Engineering from College of Management and Economics at Tianjin University, Tianjin, China. Currently, he is a Professor at the School of Management, Hebei University. His research interests include supply chain network design, healthcare management, intelligent algorithms, uncertain optimization and data-driven optimization. He has authored or coauthored over 40 articles on those areas in journals such as *IJPE*, *IJPR*, *ANOR*, *ITOR*, *JORS*.

Jianmin Xu received the B.S. and M.S. degrees from the Department of Cyberspace Security & Computer at Hebei University, Baoding, China. He obtained the Ph.D. degree in Information Management and Information Systems from the College of Management and Economics at Tianjin University, Tianjin, China. Currently, he is a Professor at the College of Cyberspace Security & Computer at Hebei University, Baoding, China. His current research interests are in information retrieval and data mining, topic discovery and tracking, and social network.